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REVIEW

A Comprehensive Review of Complex Logical Reasoning in Large Vision-Language Models

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ABSTRACT: Large Vision-Language Models (LVLMs) have achieved strong performance in multimodal perception, understanding, and generation, but their ability to perform complex logical reasoning remains insufficiently understood. In particular, it is still unclear whether current LVLMs can reliably conduct explicit logical operations, multi-step inference, abstract relational reasoning, and cross-modal evidence integration. Reasoning abilities such as deductive, inductive, abductive, multi-hop, and causal inference are fundamental to robust decision making, trustworthy interaction, and real-world deployment, yet they have not been systematically examined in the LVLM literature. Existing surveys mainly discuss mathematical reasoning, general multimodal intelligence, or benchmark progress, but they do not provide a unified account of complex logical reasoning in LVLMs, including its definition, reasoning types, modeling paradigms, evaluation protocols, and unresolved limitations. To address this gap, this survey develops a unified analytical framework for complex logical reasoning in LVLMs. This survey provides a structured review of this emerging area. We first formalize complex logical reasoning in multimodal settings and organize the literature into five recurrent reasoning families: deductive, inductive, abductive, multi-hop, and causal reasoning. We then review reasoning-oriented LVLM architectures, including unified, modular, and tool-augmented paradigms, and summarize major reasoning mechanisms such as chain-of-thought, program-based reasoning, self-correction, and interpretability-oriented analysis. We further examine representative benchmarks and evaluation protocols, with particular attention to the mismatch between final-answer accuracy and genuine reasoning validity. Based on empirical evidence from representative LVLMs and datasets, we identify common capability trends, recurring failure modes, and key open challenges. Our analysis shows that current LVLMs still struggle with reasoning faithfulness, long-horizon inference, cross-modal grounding, hallucination control, and process-aware evaluation. Finally, we outline future directions in reasoning-oriented data construction, model design, training strategies, evaluation methodology, and deployment. Overall, this survey offers a unified conceptual framework and technical roadmap for advancing LVLMs from strong perceptual systems toward reliable multimodal reasoning agents.

KEYWORDS: Large vision-language model; complex logical reasoning; multimodal reasoning; chain-of-thought; evaluation benchmark; reasoning faithfulness

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1 Introduction

In recent years, artificial intelligence has witnessed rapid progress in large language models (LLMs) and their multimodal extensions [1–3]. Large Vision-Language Models (LVLMs) [4], which integrate visual perception with language understanding and generation, have demonstrated unprecedented performance in various multimodal tasks such as image captioning, image-text dialogue, and visual question answering (VQA) [5–8]. Fig. 1 provides an overview of this transition from multimodal perception to complex logical reasoning and summarizes the survey roadmap.

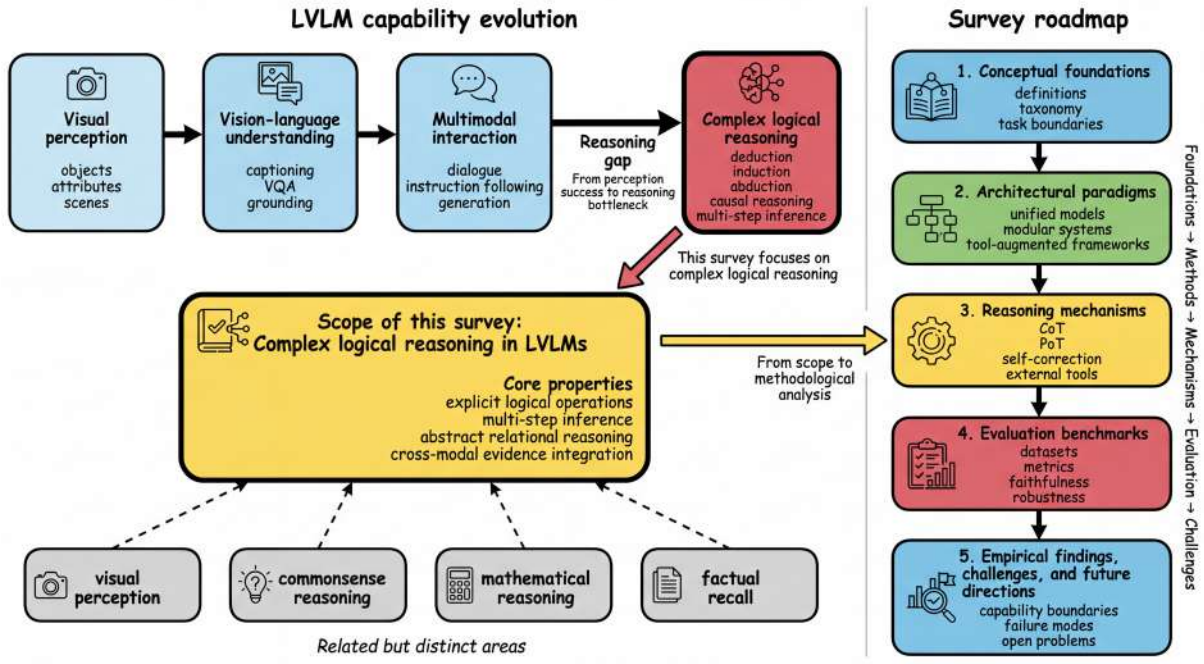


Figure 1: From multimodal perception to complex logical reasoning in LVLMs: an overview of the capability transition, the scope of this survey, and the roadmap of our review.

Complex logical reasoning is essential for moving LVLMs beyond perception-oriented tasks toward reliable multimodal agents [1,2]. In practical settings, models must often infer hidden relations, combine distributed evidence, explain causal events, resolve inconsistencies between images and text, and justify why a conclusion follows from the available evidence. The same challenge extends to dynamic multimodal scenarios, where event ordering or audiovisual cues may also need to be interpreted coherently [9,10].

Accordingly, the central problem addressed in this survey is the gap between LVLMs' strong multimodal perception and their still limited ability to perform reliable complex logical reasoning. In this paper, we use the term to denote multimodal inference that requires explicit transformation of premises, maintenance of multi-step dependencies, or composition of evidence across sources, rather than direct perceptual lookup or simple retrieval. While modern LVLMs excel at perceptual tasks such as object recognition and scene description, this limitation is especially concerning because reasoning quality directly affects high-stakes applications such as medical diagnosis, legal analysis, scientific discovery, and complex decision-making [11–13].

In medical imaging, for example, LVLMs may need to combine visual findings with textual clinical context; in autonomous driving, they must connect object states, scene dynamics, and safety constraints before acting [12,14].

Despite rapid progress, research on complex logical reasoning in LVLMs remains fragmented [2,3,15]. Previous reviews have discussed mathematical reasoning or broad multimodal capability, but they do not provide a focused treatment of logical reasoning as a distinct multimodal research problem [16–20]. As a result, three gaps remain especially prominent: the concept itself is still ambiguously defined; current benchmarks emphasize final-answer accuracy more than reasoning faithfulness; and advances in architectures, prompting, and evaluation are often reported in isolated settings that are difficult to compare systematically [21–24]. Accordingly, the survey is organized in a problem-to-evidence order: we first operationalize the target reasoning scope, then compare architectures and reasoning mechanisms under a common framework, and finally synthesize benchmark limitations, empirical trends, and unresolved bottlenecks.

1.1 Research Motivation

This survey is motivated by three specific concerns, detailed below:

(1) The definition of *complex logical reasoning* in multimodal settings is often ambiguous, with unclear boundaries relative to perceptual understanding, commonsense reasoning, and mathematical problem solving [11,25,26].

(2) Meanwhile, existing research lacks a unified evaluation framework. Current benchmarks and methodologies are scattered and typically focus on final-answer accuracy, such as OlympiadBench [27] and TheoremQA [28], offering limited insight into reasoning faithfulness and intermediate inference processes [29–31].

(3) In addition, this field is rapidly evolving, with architectural innovations, prompting strategies, reasoning mechanisms, and evaluation protocols emerging at an accelerating pace [12,32]. However, these techniques are frequently assessed in isolated settings, making systematic comparison across methods difficult [24,29].

To address these issues, this survey focuses specifically on *complex logical reasoning* in LVLMs, distinguishing it from related areas such as mathematical reasoning, commonsense reasoning, and visual perception [11,26,33]. We define it as multimodal reasoning in which the final answer cannot be obtained by direct perceptual lookup alone, but instead requires at least one inferential transformation such as explicit logical operations, multi-step dependency tracking, abstract relation handling, or cross-source evidence composition [10,23,34–36]. We primarily cover work published from 2022 to early 2026, while retaining foundational earlier studies for historical context [22,33,37–39]. Our survey covers five primary dimensions: conceptual foundations, architectural paradigms, reasoning mechanisms, evaluation benchmarks, and empirical findings, along with discussions of current challenges and future directions [12,24,25].

The main contributions of this survey are summarized as follows:

1. **Problem formalization and conceptual taxonomy.** We clarify the scope of complex logical reasoning in LVLMs and distinguish it from visual perception, commonsense reasoning, mathematical reasoning, and factual recall. We further organize existing studies into five recurrent reasoning families under a hierarchical taxonomy: deductive, inductive, abductive, multi-hop, and causal reasoning [10,21,23,34,35].
2. **Structured methodological analysis.** We analyze reasoning-oriented LVLM architectures, including unified, modular, and tool-augmented paradigms [40–42], and review major reasoning mechanisms

such as chain-of-thought, program-based reasoning, self-correction, and interpretability-oriented analysis [1,43,44]. For representative methods, we highlight their methodological focus, typical application scenarios, advantages, and limitations to facilitate comparison across different lines of work.

3. **Critical evaluation analysis.** We present a comprehensive overview of benchmarks and evaluation protocols for complex logical reasoning in LVLMs, including existing datasets, evaluation metrics, and their limitations [30,31,45]. We compare the characteristics of different benchmarks, highlighting gaps in current evaluation methodologies and opportunities for future benchmark development [29,46]. In particular, we summarize commonly used evaluation criteria, such as answer accuracy, reasoning consistency, robustness, and process-level faithfulness, so that readers can better understand the utility and limitations of different datasets and metrics.
4. **Empirical analysis.** We synthesize representative empirical findings to reveal recurring behavior patterns, capability boundaries, and common failure modes [47–49]. We further analyze factors that contribute to reasoning success and failure to guide future developments [25,49].
5. **Challenges and forward-looking research roadmap.** We identify several major challenges hindering the advancement of complex logical reasoning in LVLMs and propose future research directions across various dimensions [11,12,24].

In addition to the above contributions, this survey also provides extensive references to foundational and state-of-the-art works, serving as a comprehensive resource for researchers entering the field [2,3,50]. We organize our discussion around clear conceptual distinctions and provide examples drawn from recent literature to illustrate key points [12,24,25].

1.2 Literature Search and Screening Methodology

We conducted a structured search across major academic and open-access platforms, including Google Scholar[1], arXiv[2], IEEE Xplore[3], ACM Digital Library[4], SpringerLink[5], ScienceDirect[6], OpenReview[7], and the ACL Anthology[8]. To improve coverage, we also performed backward snowballing from key survey and benchmark papers. The literature search for the present revision was last updated on 30 May 2026. The search strategy combined keywords related to LVLMs and reasoning, such as "large vision-language model", "multimodal reasoning", "logical reasoning", "chain-of-thought", and specific reasoning types (e.g., deductive, inductive, causal). These were used both individually and in combination (e.g., "LVLM AND logical reasoning"). We applied the following inclusion criteria: (1) studies on LVLMs or multimodal reasoning systems; (2) work addressing reasoning mechanisms, architectures, benchmarks, evaluation, or empirical failure analysis; (3) peer-reviewed papers or publicly available preprints with sufficient technical detail; and (4) clear relevance to complex logical reasoning. We focused primarily on works published between 2022 and early 2026, while retaining a limited number of earlier foundational studies. Exclusion criteria included duplicate records, unimodal papers without direct multimodal transfer relevance, multimodal papers focused only on captioning or retrieval without reasoning analysis, leaderboard-style reports lacking methodological detail, and items for which the full technical content could not be examined.

[1] Google Scholar: https://scholar.google.com
[2] ArXiv Platform: https://arxiv.org
[3] IEEE Xplore: https://ieeexplore.ieee.org
[4] ACM Digital Library: https://dl.acm.org
[5] SpringerLink: https://link.springer.com
[6] ScienceDirect: https://www.sciencedirect.com
[7] OpenReview: https://openreview.net
[8] ACL Anthology: https://aclanthology.org

The screening process consisted of three stages: (1) initial collection via database search and reference expansion; (2) title and abstract filtering using the above inclusion/exclusion rules; and (3) full-text review for technical relevance, contribution type, and fit to one or more survey dimensions. The retained studies were then organized into conceptual/foundational works, architecture-oriented works, reasoning-mechanism works, and benchmark/evaluation works. Because this survey was developed as a structured narrative review rather than a preregistered systematic review, and because the earliest candidate pool was expanded iteratively during drafting and backward snowballing, exact stage-wise exclusion rates were not preserved in a form suitable for retrospective reporting. We therefore report the final corpus composition as a transparency summary rather than claiming PRISMA-style stage counts for the earliest drafting stage.

Following this process, the retained corpus comprises 138 final cited references and spans international conferences, journal publications, and a large proportion of preprints, reflecting the fast-evolving nature of the field. Most works were published between 2023 and 2025.

For greater quantitative transparency, we further summarize the retained corpus here. The current manuscript cites 138 unique works in its final reference list. The cited corpus is strongly recent: 101 of the 138 cited works (73.2%) were published between 2023 and 2025. To summarize topical coverage without forcing each paper into a single exclusive bin, we report overlapping section-level counts: 58 architecture-related citations, 42 reasoning-mechanism citations, and 56 benchmark/evaluation citations, in addition to conceptual and survey papers used for problem framing. This composition improves coverage of a rapidly moving field, but it also introduces clear coverage biases: the survey favors recent English-language public papers, benchmark-heavy studies, and open preprints, while closed industrial evaluations, negative results, and papers with limited methodological disclosure are more likely to be underrepresented.

1.3 Content Organization

The remainder of this survey is organized as follows. Section 2 introduces conceptual foundations of complex logical reasoning in LVLMs and presents the proposed taxonomy. Section 3 examines LVLM architectures designed for logical reasoning. Section 4 reviews reasoning mechanisms and strategies. Section 5 summarizes benchmarks and evaluation protocols Section 6 discusses challenges and open problems. Section 7 outlines future research directions. Section 8 concludes the survey.

2 Conceptual Foundations: Complex Logical Reasoning in LVLMs

Understanding complex logical reasoning in the context of LVLMs requires precise definitions and clear distinctions from related concepts. This section establishes the conceptual foundations by defining what constitutes logical reasoning in multimodal contexts and presenting a comprehensive taxonomy of reasoning types. Our aim is to provide a rigorous conceptual framework that can guide both research and evaluation in this rapidly evolving field [24,25].

The study of complex logical reasoning in LVLMs draws upon multiple disciplines, including philosophy (particularly logic and epistemology), cognitive science (studying human reasoning processes), and artificial intelligence (formal methods for automated inference). Furthermore, Zhou et al. [19] suggest that reasoning performance depends not only on model architecture but also on training strategy, reward design, and trajectory optimization. By integrating perspectives from these fields, we can develop a more complete understanding of what complex logical reasoning means in multimodal contexts and what challenges must be addressed to advance LVLM capabilities [21,51].

2.1 Defining Complex Logical Reasoning in Multimodal Contexts

Complex logical reasoning in LVLMs differs fundamentally from pure textual reasoning and simple visual perception. The core distinction lies between perceptual-level and logical-reasoning-level understanding. Perceptual-level understanding involves recognizing objects, identifying attributes, and describing scenes, tasks that modern LVLMs perform with remarkable accuracy [5,40,52,53]. In contrast, complex logical reasoning requires manipulating abstract representations, applying inference rules, and deriving conclusions not explicitly given in the input [1,21,23]. Fig. 2 illustrates this transition from perceptual recognition to higher-level multimodal inference.

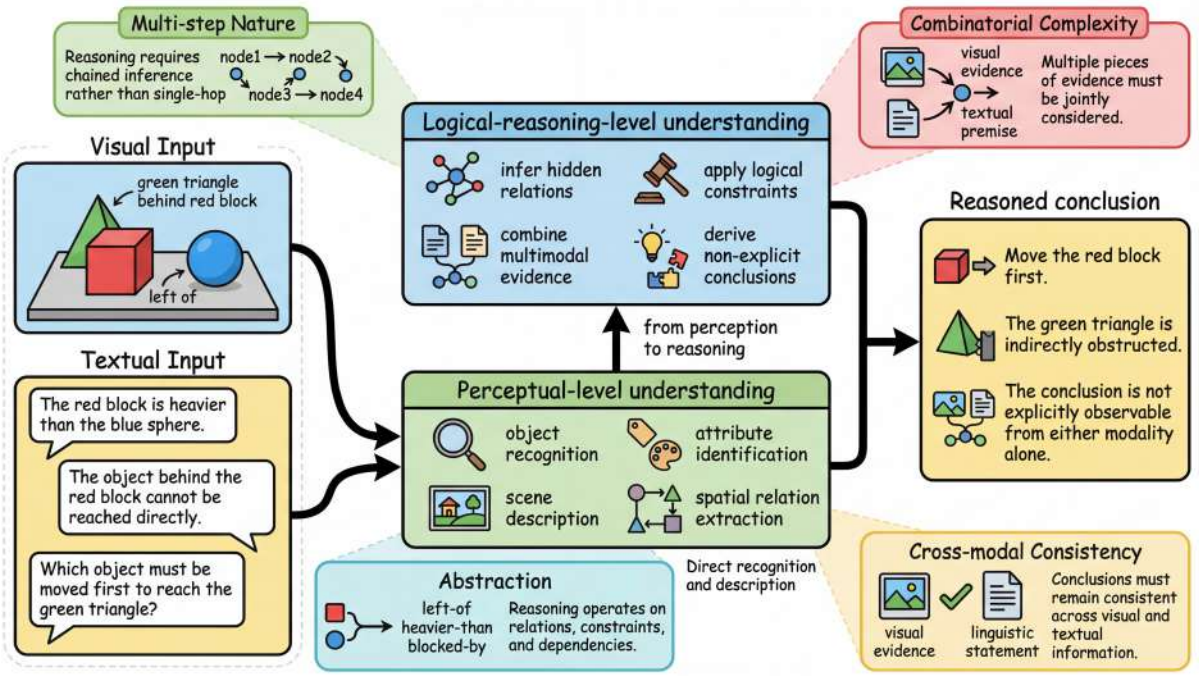


Figure 2: From perceptual understanding to complex logical reasoning in multimodal contexts. Multimodal inputs are first processed at the perceptual level to recognize objects, attributes, and basic relations, and are then transformed into higher-level reasoning representations that support multi-step inference and implicit conclusion generation.

The distinction between perceptual and complex logical reasoning can also be understood through the lens of cognitive hierarchies. Perceptual cognition operates on sensory data to extract features, objects, and basic relations. Logical cognition operates on these extracted representations to reason about implications, dependencies, and abstract relationships. While perception is often relatively direct for both humans and AI systems, complex logical reasoning typically requires deliberate multi-step inference and may rely on explicit or implicit logical rules [11,25].

Complex logical reasoning in LVLM contexts exhibits several key characteristics that distinguish it from simpler forms of multimodal understanding:

1. **Multi-Step Nature**: Complex logical reasoning typically requires chaining multiple inference steps, where each step builds upon previous conclusions. This property distinguishes logical reasoning from single-hop inference, where conclusions can be derived directly from premises without intermediate steps. Multi-step reasoning is essential for tasks such as solving complex puzzles, analyzing scenarios

with multiple constraints, and answering questions that require integrating information from different sources [1,54–57].

2. **Combinatorial Mathematicality**: Reasoning often involves combining information from multiple sources or modalities, requiring the integration of visual evidence with textual premises. This combinatorial aspect is particularly challenging because it requires maintaining consistency across different representational formats and ensuring that conclusions properly account for all available evidence. For example, a reasoning task may require combining visual evidence about object positions with textual premises about their relationships in order to infer their interactions [36,58].
3. **Abstraction**: Complex logical reasoning operates on abstract representations rather than concrete perceptual features, involving concepts such as relations, dependencies, and constraints. Abstraction allows reasoning to generalize across specific instances and apply general rules to novel situations. In LVLMs, this requires moving beyond pixel-level representations toward more abstract representations that capture the semantic and logical structure of scenes [11,22,46].
4. **Cross-Modal Consistency**: In multimodal settings, complex logical reasoning requires maintaining consistency across visual and linguistic representations, ensuring that conclusions drawn from visual evidence align with textual information. This requirement is particularly challenging because different modalities may provide complementary, partially overlapping, or even conflicting information. Reasoning systems must therefore detect and resolve such conflicts to produce valid conclusions [12,59–61].

Operationally, this survey treats a task as *complex logical reasoning* only when the final answer cannot be obtained through a single perceptual lookup or plain evidence retrieval, but instead requires at least one non-trivial inferential transformation over multimodal evidence. Such transformation may take the form of rule application, hypothesis selection, causal attribution, constraint satisfaction, or cross-source composition. Under this criterion, multi-step length alone is not sufficient: a retrieval pipeline that merely gathers several relevant facts without comparing, composing, or logically constraining them is not counted as complex logical reasoning in our taxonomy. By contrast, a multi-hop setting falls within scope when intermediate evidence must be integrated into a new conclusion whose validity depends on the consistency of the whole inference chain [54–56].

2.2 Main Types of Complex Logical Reasoning in LVLM Research

Drawing upon established frameworks from cognitive science and logic, we organize the literature into five recurrent reasoning families: deductive, inductive, abductive, multi-hop, and causal reasoning [11,25]. We do not treat these families as a strictly flat or mutually exclusive partition. Instead, we use a hierarchical reading of the taxonomy: deductive, inductive, abductive, and causal reasoning are *operator-centered* categories defined by the dominant inferential goal, whereas multi-hop reasoning is a *structural* category indicating that evidence or sub-goals must be chained across several steps. This framing reflects actual LVLM research more faithfully, because a model may be abductive and multi-hop at the same time, or causal and multi-hop, without creating a contradiction in the classification [24]. Fig. 3 summarizes this hierarchical organization of reasoning families.

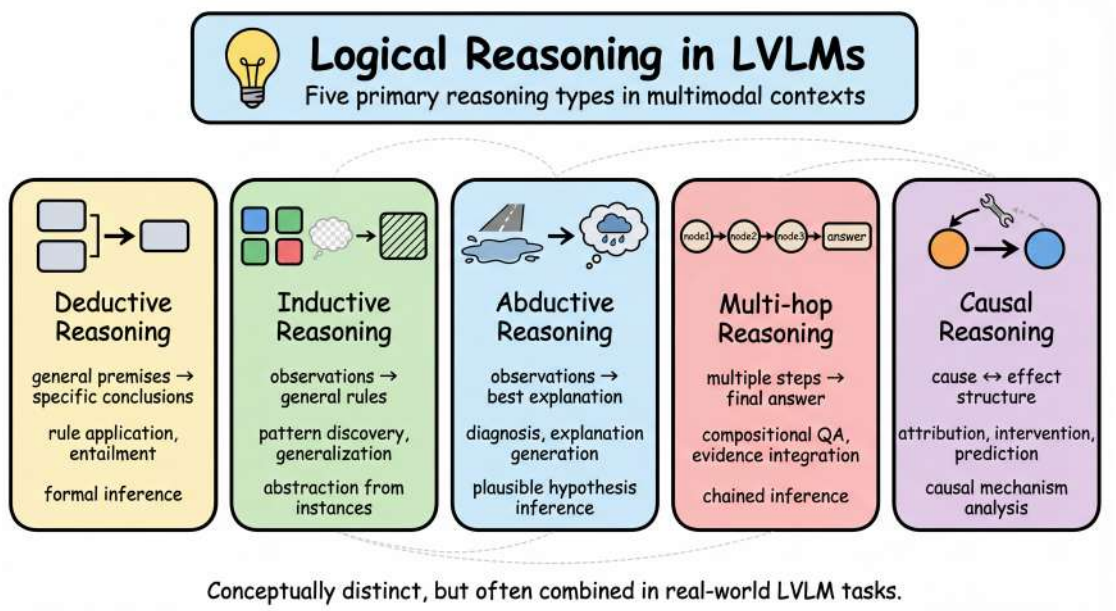


Figure 3: Taxonomy of the main reasoning families studied in LVLM research. Deductive, inductive, abductive, and causal reasoning correspond to dominant inference operators, whereas multi-hop reasoning denotes a cross-cutting structural pattern in which evidence must be chained across steps.

To improve the comparability of related studies, Table 1 links each family to its operational definition and typical task settings. We assign representative works according to their *dominant inference objective*: formal entailment for deductive reasoning, rule abstraction for inductive reasoning, explanation generation for abductive reasoning, intervention or cause-effect analysis for causal reasoning, and cross-step evidence chaining for multi-hop reasoning. When a work clearly spans multiple families, we use its main evaluation target as the primary label and discuss the overlap explicitly in the surrounding text.[9] The subsequent architecture, mechanism, and benchmark sections further discuss the methodological strengths, application scenarios, evaluation metrics, and limitations of these representative works.

Table 1: Hierarchical taxonomy of complex logical reasoning in LVLM research. Deductive, inductive, abductive, and causal reasoning are operator-centered families, while multi-hop reasoning is a structural family. Representative works are categorized by their dominant inference objective.

Reasoning Type	Definition	Typical Tasks	Representative Works
Deductive Reasoning	Deriving specific conclusions from general premises through formal logic rules	Rule application, premise validation, logical entailment	LINC [62], NS-CL [63], ReClor [64], R1-Onevision [65]

(Continued)

[9]The representative works are categorized by their dominant inference objective rather than by an assumption of mutual exclusivity. In particular, multi-hop reasoning is treated as a cross-cutting structural pattern that may co-occur with deductive, abductive, or causal reasoning.

Table 1 (continued)

Reasoning Type	Definition	Typical Tasks	Representative Works
Inductive Reasoning	Abstracting general patterns or rules from specific instances	Pattern discovery, rule induction, generalization	RAVEN [22], IndMKG [66], InPhyRe [67], WeThink [68], Molmo [69]
Abductive Reasoning	Inferring the most plausible explanation for observed phenomena	Cause inference, explanation generation, diagnosis	Reasoner [23], MAR [9], Vision-R1 [70], WeThink [68]
Multi-hop Reasoning	Chaining evidence or sub-goals across multiple steps; may co-occur with deductive, abductive, or causal operators	Composite question answering, cross-modality inference	BRIDGE [55], M3GQA [54], MultiHop-RAG [56], Chameleon [71], LLaVA-NeXT-Interleave [72], Qwen2.5-VL [73]
Causal Reasoning	Identifying causal relationships between events or variables	Causal attribution, effect prediction, intervention analysis	CausalVLBench [34], CausalBench [35], MuCR [10], VcCoT [10], Reasoner [23], Vision-R1 [70]

Deductive Reasoning. Deductive reasoning involves deriving specific conclusions from general premises through the application of formal logical rules [21,51]. In LVLM contexts, deductive reasoning typically appears in tasks requiring rule application, premise validation, or logical entailment determination [64]. For example, given a set of premises about relationships between objects and a query about a specific relationship, deductive reasoning enables the model to infer the answer by applying the relevant rule structure.

The formal nature of deductive reasoning makes it particularly amenable to neuro-symbolic approaches that combine neural representation learning with symbolic inference engines [41,51]. Works such as ReClor [64] have developed benchmarks specifically targeting deductive reasoning capabilities in language models. ReClor presents reading comprehension questions that require complex logical reasoning to determine entailment or contradiction relationships between premises and conclusions.

A key challenge in deductive reasoning for LVLMs lies in handling the semantic gap between natural language premises and formal logical representations [62]. Natural language is often ambiguous and may not map cleanly to formal logical forms. Furthermore, real-world premises may be incomplete or may contain implicit assumptions that must be identified for valid deduction.

Deductive reasoning can be further subdivided into propositional deduction (reasoning about the truth values of propositions) and predicate-logic deduction (reasoning about object properties and relations). In LVLM contexts, both forms are relevant, with predicate-level deduction being particularly important for reasoning about visual scenes that involve multiple objects and their interactions [74].

Inductive Reasoning. Inductive reasoning involves discovering general patterns or rules from specific instances or observations [11]. This form of reasoning is fundamental to scientific discovery and learning from experience. In LVLM contexts, inductive reasoning tasks include completing visual patterns, generalizing from examples, and identifying underlying rules.

The RAVEN benchmark [22] represents a significant contribution to evaluating inductive reasoning in visual contexts, presenting matrices in which models must identify underlying transformation rules. RAVEN includes visual matrices with varied structures, requiring models to understand relations between elements and apply transformation rules to complete the pattern.

Inductive reasoning in visual contexts presents unique challenges due to the complexity of visual representations and the need to handle variations in style, viewpoint, and context [46]. Unlike formal inductive reasoning in logic, visual induction often requires dealing with noisy, high-dimensional input and may involve probabilistic rather than deterministic generalization.

Inductive reasoning can be characterized by several key properties: the resulting generalization is uncertain (the conclusion may be false even if the observations are correct), the reasoning moves from specific observations to broader rules, and the strength of the generalization depends on the representativeness of the observed instances [11]. These properties make inductive reasoning fundamentally different from deductive reasoning and require different computational approaches.

Abductive Reasoning. Abductive reasoning, also known as inference to the best explanation, involves inferring the most plausible explanation for observed phenomena [9,23,75]. In LVLM contexts, abductive reasoning is particularly relevant for tasks such as visual explanation generation, where models must infer what events or conditions could have produced an observed scene.

Recent works such as KN-VLM [76] have focused on visual abductive reasoning, developing models that can infer hypotheses to explain visual contexts. This capability is essential for applications such as accident analysis and safety monitoring, where systems must infer what events could have led to an observed situation.

Abductive reasoning is particularly challenging because it requires both understanding observed phenomena and generating plausible explanations that account for them [23]. Unlike deductive reasoning, where conclusions follow necessarily from premises, abductive reasoning involves selecting the most plausible explanation among multiple candidates, often under incomplete or uncertain information.

Multi-Hop Reasoning. Multi-hop reasoning refers to the structural requirement that evidence or subgoals be chained across multiple steps [54,56,57]. In this survey, it is treated as a cross-cutting reasoning pattern rather than a purely separate operator on par with deduction or abduction. A task is multi-hop when the model must maintain intermediate states and compose dispersed evidence before the final answer can be derived.

The CLEVR benchmark [39] represents a foundational contribution to multi-hop reasoning evaluation in visual question answering. CLEVR presents synthetic scenes with multiple objects and asks questions that require multi-step reasoning to answer. For example, a question might ask, "What is the color of the object that is to the left of the red sphere?", which requires first identifying the red sphere, then finding objects to its left, and finally extracting the relevant color attribute.

This distinction is important for separating reasoning from simple retrieval. If a system merely fetches several relevant snippets or regions and one of them directly contains the answer, the process is multi-source retrieval but not necessarily multi-hop reasoning. By contrast, reasoning is required when retrieved pieces must be compared, composed, or constrained to infer a conclusion that is not explicitly stated in any single source. Multi-hop reasoning is therefore particularly relevant for real-world applications in which questions require integrating information across different parts of an image or across multiple images [36]. The ability

to maintain intermediate conclusions and chain reasoning steps is essential for complex question answering, narrative understanding, and comprehensive scene analysis.

Causal Reasoning. Causal reasoning involves identifying causal relationships between events or variables, distinguishing correlation from causation, and predicting the effects of interventions [10,34,35]. This form of reasoning is essential for understanding how changes in one variable affect others and for predicting the outcomes of actions.

Recent works such as the Multimodal Causal Reasoning Benchmark [10] have developed benchmarks specifically targeting causal reasoning in multimodal contexts. These benchmarks challenge LVLMs to identify causal links across modalities, requiring the joint understanding of visual and textual information.

Causal reasoning is fundamental to many real-world applications, such as autonomous driving (predicting the effects of actions) [14]. The ability to distinguish causal relationships from simple correlations is a key capability that separates robust reasoning systems from superficial pattern-matching systems.

Causal reasoning in LVLMs faces unique challenges due to the need to handle both visual and textual information about causal relationships [10,34,35]. Visual information may show how events unfold over time, while textual information may describe causal mechanisms or provide explanatory context. Integrating these modalities to perform causal reasoning requires sophisticated representation learning and inference capabilities [10].

2.3 Conceptual Boundaries of Complex Logical Reasoning

To avoid ambiguity in the subsequent taxonomy, we first clarify the conceptual boundaries between complex logical reasoning and related paradigms. Despite the increasing use of the term *complex logical reasoning* in LVLM research, its conceptual boundaries remain ambiguous and often overlap with related notions such as commonsense reasoning, mathematical reasoning, and visual perception [26,33,77]. This ambiguity may lead to inconsistent evaluation and weaken the validity of taxonomic frameworks. To address this issue, we provide a structured clarification of how complex logical reasoning differs from, and interacts with, these closely related paradigms.

We define *complex logical reasoning* as the ability to perform multi-step, structured inference over multimodal inputs, where intermediate reasoning steps are required to derive the final answer [1,25]. More specifically, a task falls into this scope when it requires at least one inferential transformation beyond direct lookup, such as rule application, evidence composition, contradiction resolution, or hypothesis comparison. This distinguishes it from *perception*, which primarily involves recognizing or extracting information from visual inputs (e.g., object detection or attribute identification), and from *commonsense reasoning*, which relies on implicit world knowledge and prior experience rather than explicit multi-step deduction [33]. It also clarifies the relation to *compositional reasoning*: compositional structure is often a useful ingredient, but it becomes complex logical reasoning only when composed parts must support a new inferential conclusion rather than a direct structural readout. In contrast, *mathematical reasoning* represents a more formalized and symbolic subset of logical reasoning, often involving precise calculations, algebraic manipulation, or rule-based derivations [15,26]. While mathematical reasoning is inherently logical, it typically operates in a more constrained and well-defined problem space compared to general multimodal reasoning [78,79].

Importantly, these paradigms are not mutually exclusive but instead exhibit hierarchical and compositional relationships. Complex logical reasoning often builds upon perceptual grounding and commonsense knowledge, while also incorporating mathematical reasoning in specialized tasks [11,13]. For example, solving a visual math problem may require perception (reading a diagram), commonsense (interpreting context), and symbolic reasoning (performing calculations) [80,81]. Therefore, rather than treating these categories

as strictly separate, we view complex logical reasoning as an integrative capability that coordinates multiple cognitive components [12,77]. Fig. 4 further illustrates these conceptual boundaries by contrasting their input focus, core operation, evidence type, and typical failure mode.

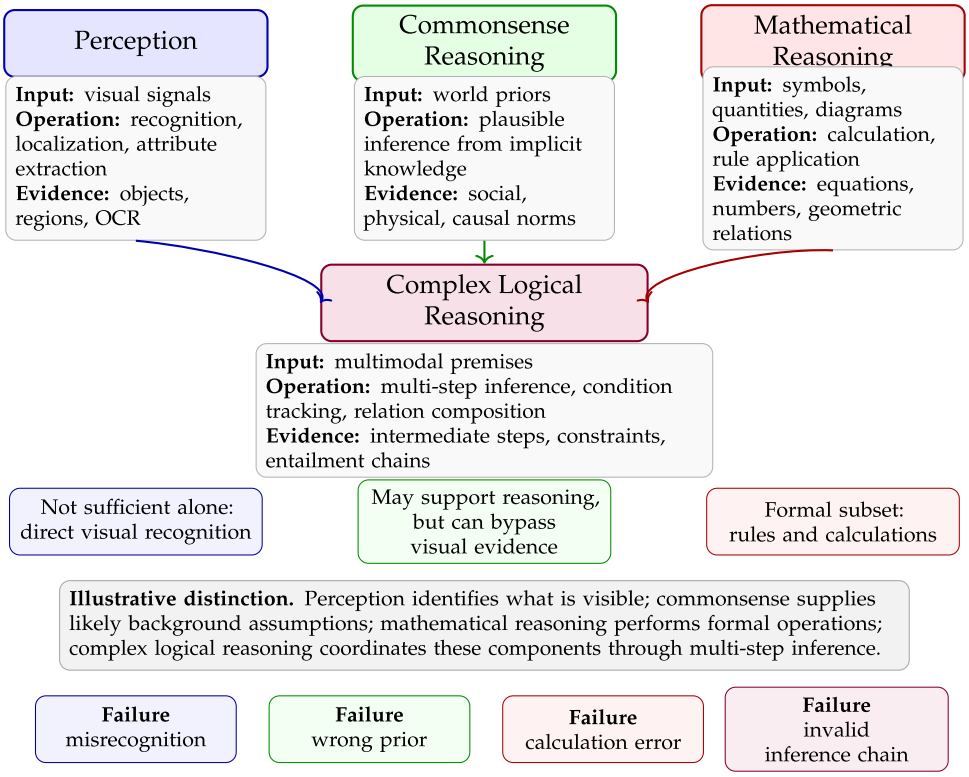


Figure 4: Conceptual boundaries among perception, commonsense reasoning, mathematical reasoning, and complex logical reasoning in LVLM research. Complex logical reasoning is distinguished by its dependence on multi-step inference over multimodal premises, although it may incorporate perceptual grounding, commonsense priors, and mathematical operations as supporting components.

2.4 Relationship with Other Reasoning Paradigms

As shown in Fig. 5, complex logical reasoning in LVLMs is closely related to several other reasoning paradigms, including commonsense, temporal, spatial, counterfactual, and strategic reasoning [24,25]. Although these paradigms emphasize different aspects of inference, they frequently interact in practical multimodal tasks. Complex logical reasoning provides a structured inference framework, while the other paradigms contribute complementary capabilities such as background knowledge, temporal ordering, spatial constraint modeling, hypothetical analysis, and decision-oriented planning. Understanding these relationships is important for developing comprehensive reasoning systems that can handle the full range of real-world applications.

Commonsense Reasoning. Commonsense reasoning involves applying everyday knowledge about the world to interpret situations and make reasonable inferences [33]. It frequently overlaps with logical reasoning, particularly abductive reasoning [9,23], when explaining observed phenomena. For example, if we see a wet sidewalk and dark clouds in the sky, commonsense reasoning allows us to infer that it likely rained.

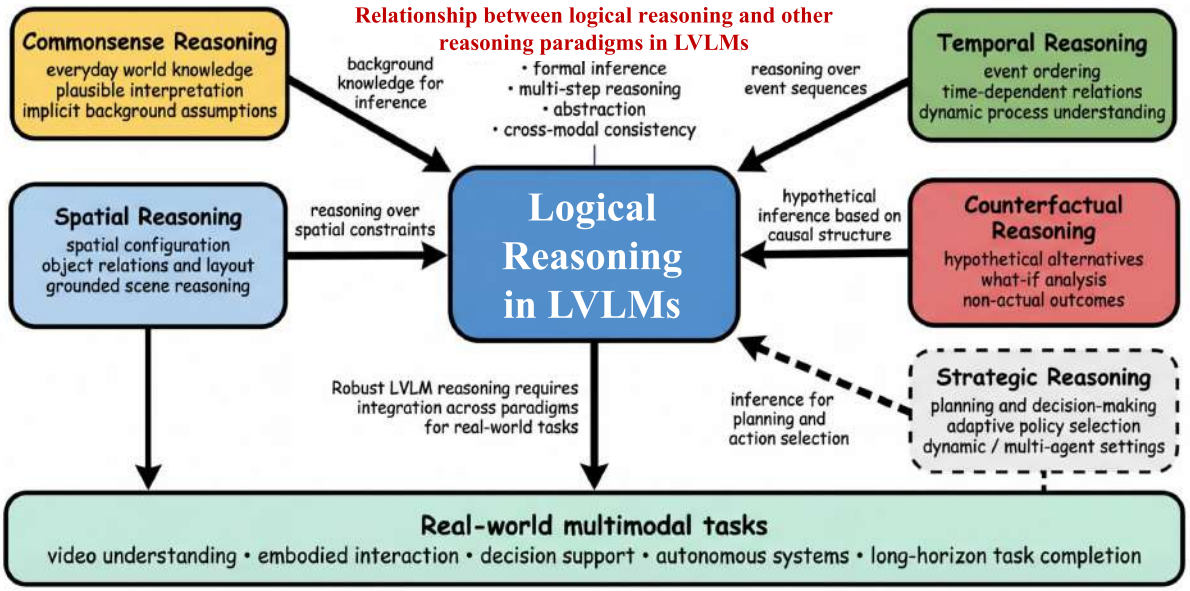


Figure 5: Relationship between logical reasoning and other reasoning paradigms in LVLMs. Logical reasoning serves as a core capability for multimodal inference, but it is closely connected with several related paradigms, including commonsense, temporal, spatial, counterfactual, and strategic reasoning.

Commonsense reasoning differs from formal logical reasoning in that it relies on implicit knowledge about the world rather than explicit logical rules [25]. This implicit knowledge is difficult to formalize and may vary across cultures and contexts. In LVLM contexts, commonsense reasoning is essential for interpreting ambiguous visual scenes and making reasonable assumptions about unobserved elements [82].

The relationship between commonsense reasoning and logical reasoning is complementary: logical reasoning provides formal frameworks for inference, while commonsense reasoning provides the background knowledge needed to apply those frameworks appropriately [24]. Integrating these reasoning types remains a key challenge for LVLM development.

Temporal Reasoning. Temporal reasoning focuses on understanding time-ordered sequences of events and reasoning about time-dependent relationships [83,84]. In multimodal contexts, temporal reasoning is essential for understanding video content, narrative descriptions, and cause-effect relationships that unfold over time.

Temporal reasoning in LVLMs requires representing time and reasoning about temporal relations such as before, after, simultaneous, and during [85]. This capability is essential for tasks such as video question answering, where questions may concern the order of events or the duration of activities. Beyond explicitly reasoning-oriented video benchmarks, large-scale video understanding resources such as YouTube-8M [86] have also played an important enabling role by providing millions of annotated videos for scalable representation learning.

The integration of temporal reasoning with logical reasoning is particularly important for causal reasoning tasks, which require understanding how events unfold over time and identifying causal relationships between temporally ordered events [10,34,35].

Spatial Reasoning. Spatial reasoning deals with understanding and reasoning about physical space and spatial relationships between objects [87]. In multimodal contexts, spatial reasoning is often combined with logical deduction to answer questions about object relationships, spatial configurations, and layout understanding.

Spatial reasoning in LVLMs requires representing spatial relationships such as above, below, inside, outside, near, and far [88]. Chen et al. [89] highlights that multimodal reasoning also requires explicit referential grounding: by allowing spatial coordinates to be expressed directly in natural-language interaction, it extends MLLMs toward location-aware dialogue and grounded reasoning. GeoQA is a representative benchmark in this direction, as it explicitly evaluates geometric question answering that requires joint visual perception, diagram understanding, and numerical reasoning [81].

The integration of spatial reasoning with logical reasoning is particularly important for tasks that require reasoning about spatial constraints, such as path planning, object arrangement, and spatial inference tasks that combine visual evidence with textual premises about spatial relationships [58].

Counterfactual Reasoning. Counterfactual reasoning involves reasoning about hypothetical alternatives to past events. It requires considering what would have happened if circumstances had been different, a capability that is essential for planning, decision-making, and understanding causality [90].

Counterfactual reasoning is closely related to causal reasoning, as both involve understanding causal relationships [10,34,35]. However, counterfactual reasoning specifically focuses on hypothetical scenarios that did not actually occur, whereas causal reasoning may also concern actual causal relationships.

In LVLM contexts, counterfactual reasoning is important for tasks such as visual explanation generation, where models must consider alternative explanations for observed phenomena [23], and for decision support systems, where models must evaluate the outcomes of different actions.

These reasoning paradigms interact with logical reasoning in complex ways, and successful LVLM systems must integrate capabilities across multiple paradigms [12,24]. Realistic environments such as WebArena [91] highlight the importance of long-horizon reasoning, interaction planning, and functional task completion in complex environments. The complexity of real-world reasoning tasks often requires combining multiple reasoning types, making the development of comprehensive reasoning systems a significant challenge.

Another related paradigm is strategic reasoning, which concerns planning and decision-making under dynamic, uncertain, and often multi-agent environments. A recent survey [20] shows that strategic reasoning emphasizes anticipating others' behavior and adaptive policy selection, making it related to the logical-inference focus of many LVLM studies.

3 LVLM Architectures for Logical Reasoning

The architecture of a Large Vision-Language Model fundamentally shapes its capability to perform logical reasoning [5,6,40,92]. The choice of architecture involves trade-offs between expressiveness, training efficiency, interpretability, and the ability to handle complex reasoning tasks.

The development of LVLM architectures for logical reasoning has been driven by several key considerations. First, logical reasoning often requires maintaining intermediate conclusions and chaining reasoning steps, which poses challenges for standard sequence-to-sequence architectures [1,57]. Second, logical reasoning may benefit from explicit symbolic representations that can be verified and manipulated, which may not emerge naturally from neural architectures [41,51]. Third, the multimodal nature of LVLMs introduces challenges in aligning visual and textual representations to support reasoning across modalities [59–61].

Fig. 6 provides a unified framework view of LVLM architectures for logical reasoning. At a high level, reasoning capability is shaped by two complementary aspects: (1) the architectural paradigm that organizes perception, language understanding, and reasoning, i.e., *unified*, *modular*, *tool-augmented*, and (2) the visual feature encoding and alignment mechanisms that determine how visual evidence is represented and grounded across modalities i.e., *visual tokenization*, *cross-modal alignment*, and *spatial encoding*.

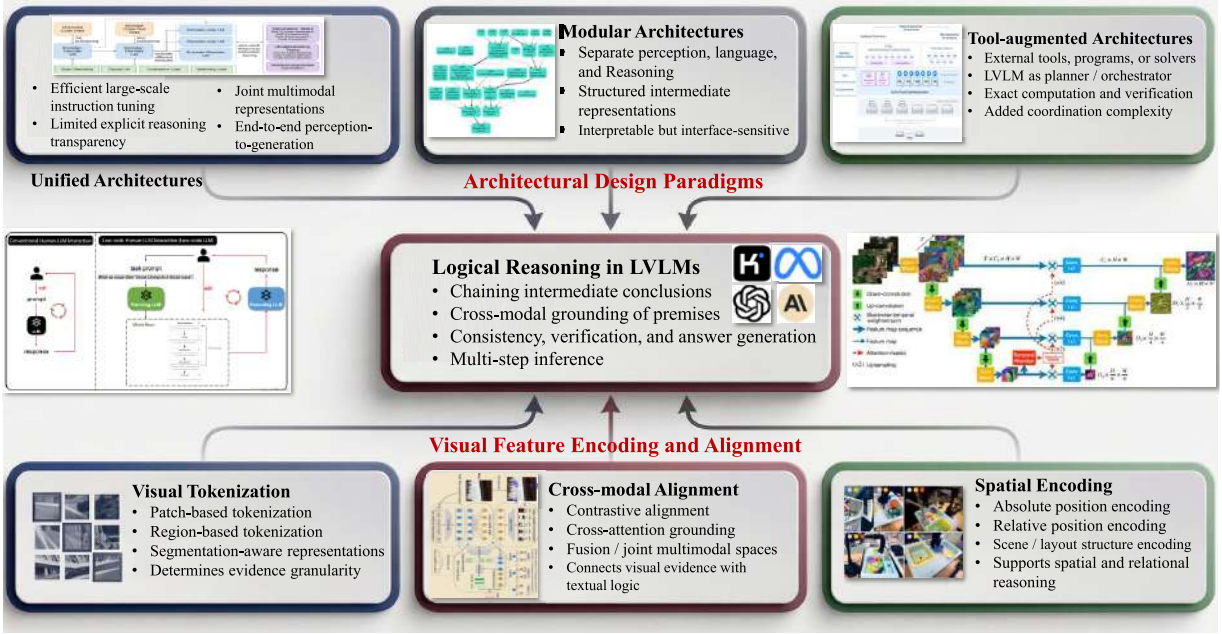


Figure 6: Framework view of LVLM architectures for logical reasoning. Upper layer: three architectural paradigms (unified, modular, tool-augmented). Lower layer: visual foundations (tokenization, cross-modal alignment, spatial encoding). Together, they shape LVLM reasoning under complex multimodal inputs.

3.1 Architectural Design Paradigms

We identify three primary architectural paradigms for logical reasoning in LVLMs: unified architectures, modular architectures, and tool-augmented architectures. Although the boundaries between these paradigms are not always strict, they reflect three distinct design philosophies for integrating visual perception, language understanding, and reasoning. Unified architectures emphasize end-to-end multimodal representation learning; modular architectures explicitly separate perception and reasoning; and tool-augmented architectures extend LVLMs with external computation or symbolic tools. This categorization provides a useful lens for comparing how different systems support logical inference under multimodal inputs. Table 2 summarizes representative architecture works under these paradigms and highlights their reasoning-oriented design choices and limitations.

Table 2: Representative LVLM architecture works for reasoning (2022–early 2026).

Model	Year	Architecture	Reasoning-Oriented Design	Limitation
Flamingo [93]	2022	Unified multimodal model with cross-attention	Frozen backbone with perceiver resampler enables few-shot multimodal in-context reasoning	Reasoning processes remain largely implicit

(Continued)

Table 2 (continued)

Model	Year	Architecture	Reasoning-Oriented Design	Limitation
BLIP-2 [40,94]	2023	Unified bridge architecture (Q-Former)	Query bottleneck improves efficient vision-language alignment for downstream reasoning	Limited explicit reasoning verification
LLaVA [5,53]	2023	Instruction-tuned LVLM	Instruction data enhances compositional reasoning and step decomposition	Susceptible to hallucination under complex reasoning tasks
InternVL [95]	2023	Scaled vision backbone with multimodal alignment	Strong visual scaling improves OCR, grounding, and long-context perception	High computational cost
MiniGPT-4 [96]	2023	Frozen vision encoder + projection + LLM	Lightweight alignment unlocks strong multimodal understanding abilities	Outputs may be fluent but not always reliably grounded
Qwen-VL [7]	2023	Unified LVLM with visual receptor and grounding-aware interface	Strengthens text reading, grounding, and multilingual multimodal reasoning	Multi-step reasoning is not the primary optimization target
Otter [97]	2023	Multimodal in-context LLM	Supports interleaved image-text instruction following and in-context reasoning	Limited robustness in complex reasoning tasks
Kosmos-2 [98]	2023	Grounded multimodal LLM with location-token linking	Integrates grounding directly into generation, supporting referring and spatially grounded reasoning	Grounded generation increases interface complexity and does not itself ensure deep logical reasoning
Qwen2-VL [99]	2024	Unified multimodal LLM with Naive Dynamic Resolution and M-RoPE	Dynamic-resolution visual tokenization and unified image/video modeling improve document, chart, and video reasoning	Primarily optimized for benchmark performance rather than interpretable reasoning processes

(Continued)

Table 2 (continued)

Model	Year	Architecture	Reasoning-Oriented Design	Limitation
LLaVA-OneVision [100]	2024	Unified multi-scenario model	Joint training across image and video reasoning settings	Limited robustness under distribution shifts
Chameleon [71]	2023	Plug-and-play LLM with external tools	Compositional reasoning with tool integration (web, vision, Python)	High complexity due to tool dependencies and planning
Reasoner/Reasoner-v2 [23]	2024	Cascaded reasoning transformer	Multi-level abductive reasoning for visual tasks	Limited applicability beyond visual abductive settings
Molmo [69]	2024	Open VLM trained with open data and curated pipeline	High-quality open pretraining/fine-tuning data improves robust multimodal reasoning without proprietary teacher models	Performance still depends strongly on data curation and scaling strategy
InternVL2.5 [101]	2024	Scaled InternVL family with model/data/test-time scaling	Explicitly benefits from CoT and test-time scaling, improving multidisciplinary and long-context reasoning	Larger-scale inference and test-time reasoning increase cost
LLaVA-NeXT-Interleave [72]	2024	Interleaved multimodal architecture for multi-image, video, and 3D	Treats interleaved multimodal context as a general template, improving cross-instance reasoning	More complex training recipe and evaluation setting
Qwen2.5-VL [73]	2025	Dynamic-resolution ViT + unified multimodal LLM	Native dynamic-resolution perception, window attention, and long-video modeling improve document, chart, GUI, and agentic reasoning	Strong answer performance, but explicit reasoning faithfulness is not always guaranteed
Vision-R1 [70]	2025	Reasoning-oriented MLLM with cold-start CoT data + RL	Uses multimodal CoT cold-start data, PTST, and GRPO to stimulate complex multimodal reasoning	Primarily optimized for reasoning benchmarks, especially math-heavy settings

(Continued)

Table 2 (continued)

Model	Year	Architecture	Reasoning-Oriented Design	Limitation
R1-OneVision [65]	2025	Cross-modal formalization pipeline + SFT/RL reasoning model	Transforms visual input into formal textual representations to enable step-by-step language-style reasoning	Formalization pipeline adds complexity and may lose raw visual nuance
WeThink [68]	2025	RL-enhanced vision-language reasoning framework	Builds a 120K reasoning-path dataset from 18 sources and applies hybrid-reward RL for general-purpose visual reasoning	Gains depend strongly on synthesized reasoning-data quality
GLM-4.1V-Think/GLM-4.5V [102]	2025	Scaled VLM family with reasoning-centric training + RLCS	Curriculum-based RL unlocks broad reasoning ability across STEM, video, grounding, GUI agents, and long documents	Training and inference recipes are comparatively heavy and complex
MMaDA [103]	2025	Unified multimodal diffusion language model	Modality-agnostic diffusion architecture, mixed long CoT fine-tuning, and UniGRPO unify reasoning and generation	Less standard than autoregressive LVLMs and harder to compare directly with mainstream VLM pipelines
VLM-R^3 [104]	2025	Region-recognition-and-reasoning framework with R-GRPO	Learns when and where to look, then injects cropped evidence back into interleaved CoT	Extra region-selection loop increases system complexity

Unified Architectures. Unified architectures process visual and textual inputs within a tightly integrated multimodal model and learn joint representations across modalities. In practice, this family includes models that rely on cross-attention connectors, alignment bridges, or unified multimodal token spaces, even when the vision encoder and language backbone are not literally a single encoder. Representative examples include Flamingo [93], BLIP-2 [40], MiniGPT-4 [96], LLaVA [5,53], and more recent systems such as Qwen2-VL [99] and Qwen2.5-VL [73].

Flamingo [93] is an early representative of this paradigm. By inserting cross-attention layers between frozen visual and language backbones, it enables multimodal few-shot in-context learning while preserving the capabilities of large pretrained components. BLIP-2 [40] further improves this design through the Q-Former, which acts as a compact interface between visual features and the language model. LLaVA [5,53] shows that visual instruction tuning can substantially improve multimodal dialogue and compositional reasoning, while MiniGPT-4 [96] suggests that part of multimodal reasoning ability can be elicited from strong language backbones through lightweight alignment alone. More recent unified models also strengthen document, chart, multi-image, and video reasoning by improving visual tokenization, long-context handling, and interleaved multimodal modeling [72,73,99,100].

The main strength of unified architectures lies in end-to-end multimodal optimization. Because perception, alignment, and generation are trained in a closely coupled manner, these models are often effective at compositional understanding and at tasks that can be acquired through large-scale multimodal instruction tuning [2,16]. However, this tight coupling also makes the reasoning process difficult to interpret or verify. Intermediate reasoning is usually implicit in hidden states rather than represented as explicit symbolic structure, which can make these models less reliable on tasks requiring strict rule application, formal verification, or faithful multi-step inference [1,44].

Modular Architectures. Modular architectures separate visual encoding, language understanding, and logical reasoning into distinct components. This separation allows each module to be optimized for its own role and makes it easier to incorporate specialized reasoning mechanisms when explicit inference is required.

A prominent instance of this paradigm is neuro-symbolic modeling, which combines neural perception with symbolic reasoning engines [41,63]. In such systems, visual and linguistic inputs are first parsed into structured intermediate representations, such as objects, relations, or logical forms. A symbolic module then performs explicit inference over these structures and produces conclusions that can be translated back into natural language. This approach is especially attractive for deductive reasoning and compositional visual reasoning because the reasoning procedure is more transparent and, in principle, verifiable [49,51].

The main advantage of modular architectures is interpretability. Because perception and reasoning are partly decoupled, one can inspect the intermediate representations and reason about where errors occur. This is particularly valuable for logical tasks in which correctness depends on precise relational structure rather than fluent answer generation. The main weakness, however, lies in the interface between continuous perceptual representations and discrete symbolic structures. Parsing errors, information loss, or misalignment between neural outputs and symbolic inputs can propagate through the pipeline and degrade reasoning quality [62,71]. As a result, modular systems may be more rigorous on structured reasoning tasks, but less flexible when dealing with ambiguous, noisy, or open-ended multimodal inputs.

Tool-Augmented Architectures. Tool-augmented architectures extend LVLMs with external tools such as code interpreters, symbolic solvers, search modules, or task-specific vision components [105,106]. Instead of requiring the core model to internalize all reasoning skills, this paradigm treats the LVLM as a coordinator that decides when to call tools, how to formulate the calls, and how to integrate the returned results.

This paradigm is particularly effective when reasoning requires forms of computation that neural models do not perform reliably on their own, such as exact calculation, formal logical checking, or structured program execution [42,43]. Representative examples include Visual Programming [107], which combines LLM-based planning with external vision modules and executable intermediate steps, and Chameleon [71], which integrates multiple external tools into compositional reasoning workflows.

The primary strength of tool-augmented architectures is flexibility. By outsourcing specialized operations to external modules, these systems can extend reasoning capabilities without redesigning the base LVLM. A deeper limitation is that tool-augmented systems shift the burden of reasoning from internal model representations to external orchestration. This introduces a new class of failure modes, including incorrect tool selection, invalid program generation, and misinterpretation of tool outputs [43,108]. In complex multi-step scenarios, such errors can accumulate across reasoning stages, making the overall system brittle despite the correctness of individual tools.

Comparative Synthesis across Paradigms. The current empirical evidence remains uneven across unified, modular, and tool-augmented architectures. Broad shared benchmark suites such as MMBench,bib31, MathVista, MuirBench, and Video-MME are currently dominated by unified or unified-style LVLMs,

which means they provide the clearest evidence for general benchmark transfer under common reporting conventions (see Section 5.3). In our standardized evidence map in Section 5.3, the systems with directly comparable results across these shared benchmarks are all unified or unified-style models, such as GPT-4V, Qwen2.5-VL, LLaVA-OneVision, and InternVL2.5 [37,73,100,101]. By contrast, modular and neuro-symbolic systems are more often evaluated on structured settings such as CLEVR, GeoQA, or targeted reasoning-faithfulness studies, where their explicit intermediate representations improve inspectability and rule-level verification but reduce direct comparability with open-ended LVLM benchmarks [39,49,81]. Tool-augmented systems report concrete gains when exact computation, symbolic execution, or external retrieval is essential, especially in program-execution or math-oriented settings, yet these gains are inseparable from the quality of orchestration and from the assumptions imposed by external tools [42,105,107]. Accordingly, current literature supports three cautious conclusions: unified architectures offer the strongest evidence of broad benchmark coverage, modular architectures offer the strongest evidence of explicit intermediate-state verification on structured tasks, and tool-augmented architectures offer the strongest evidence of exactness when external computation is reliable. None of these observations should be overgeneralized into a universal ranking, because the supporting evaluations still differ substantially in task distribution, annotation format, and end-to-end system boundary.

3.2 Visual Feature Encoding and Alignment

The quality of visual feature encoding directly affects logical reasoning capability. To support multimodal inference, visual representations must preserve information that is relevant not only for recognition, but also for abstraction, relation modeling, and cross-modal grounding [52,109]. This subsection discusses three aspects that are particularly important for reasoning-oriented LVLMs: visual tokenization, cross-modal alignment, and spatial encoding. Fig. 7 presents these encoding and alignment components as a reasoning-oriented representation pipeline.

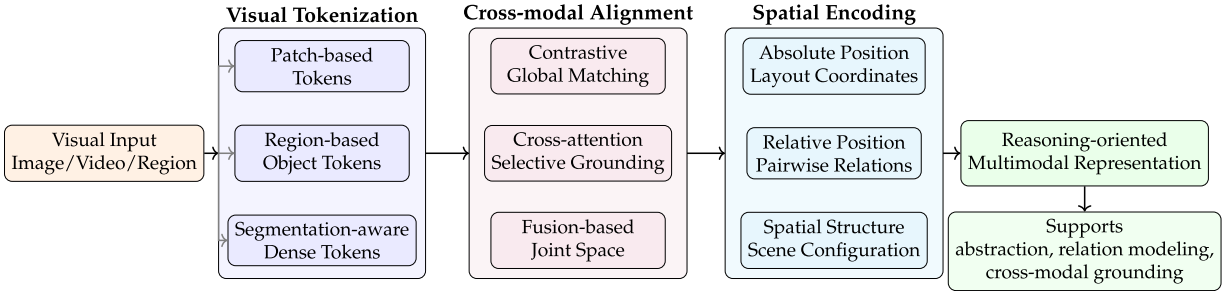


Figure 7: Visual feature encoding and alignment for reasoning-oriented LVLMs. Visual tokenization determines the granularity of visual evidence; cross-modal alignment connects visual and textual representations; and spatial encoding preserves layout, pairwise relations, and scene structure needed for logical reasoning.

Visual Tokenization. Visual tokenization determines the granularity and structure of the visual information that is made available to the language model. Different tokenization strategies support different forms of reasoning.

Patch-based tokenization, widely used in ViT-style architectures and models such as CLIP, divides an image into fixed-size patches that are treated as visual tokens [52,110]. This design is simple and scalable, and it works well for general-purpose representation learning. However, fixed patch grids do not naturally align with semantically meaningful entities, which can make object-centric and relation-centric reasoning more difficult.

Region-based tokenization instead derives features from detected objects or salient regions, making the visual representation more compatible with object-level reasoning [111,112]. This can be advantageous for tasks involving counting, comparison, and spatial or relational inference. Its limitation is that it depends on upstream detection quality and may omit contextual information that is not well captured by explicit object proposals.

A related alternative is to rely on *segmentation-aware representations*, where pixel-level or region-level semantic labels provide dense structural cues [113]. Such representations can support fine-grained reasoning about attributes and scene layout, but the predefined semantic categories may still not align well with the more abstract or relational categories required for logical reasoning.

Cross-Modal Alignment. Cross-modal alignment is the mechanism that connects visual and textual representations so that reasoning can proceed across modalities. In reasoning-oriented LVLMs, alignment must do more than capture coarse semantic similarity; it must support the binding of entities, attributes, relations, and constraints between modalities.

One common strategy is *contrastive alignment*, as used in CLIP-style pretraining, where paired image-text examples are pulled together in representation space and unpaired examples are pushed apart [52]. This produces strong global semantic representations, but it does not necessarily capture the fine-grained alignments needed for logical inference.

A second strategy is *cross-attention-based alignment*, exemplified by Flamingo [93], where the language model dynamically attends to visual features during inference. This design is more flexible and can adapt the alignment process to the current question or context, which is beneficial for reasoning tasks that require selective use of visual evidence.

A third strategy is *fusion-based integration*, in which visual and textual representations are combined through learned transformation modules or joint multimodal spaces [114]. Fusion can enable richer multimodal interaction, but if it is too coarse, it may blur modality-specific structure that remains important for reasoning.

Spatial Encoding. Spatial information is often central to logical reasoning, particularly for tasks involving object relations, geometric structure, physical layout, and scene configuration [88,115]. As a result, the way spatial structure is encoded has a direct impact on reasoning performance.

Absolute position encoding represents where tokens occur in the visual field. This can help preserve layout information, but absolute coordinates alone are often insufficient for reasoning because they do not directly capture relations between entities.

Relative position encoding instead models pairwise spatial relations such as left of, right of, above, and below. These relational encodings are generally more robust across changes in scale, viewpoint, and composition, and they are often more relevant for logical reasoning tasks.

Spatial structure encoding aims to represent higher-order layout or scene organization, enabling reasoning about configurations rather than isolated pairwise relations [74]. This is especially important for tasks involving multi-object spatial constraints, embodied interaction, or scene-level planning.

4 Reasoning Mechanisms and Strategies

The development of reasoning mechanisms in LVLMs has been strongly influenced by progress in LLM reasoning, particularly chain-of-thought prompting and its extensions, while also adapting to the additional challenges introduced by multimodal evidence [1,57]. In contrast to purely textual settings, multimodal reasoning requires not only generating intermediate inference steps, but also grounding those steps in visual inputs and maintaining consistency between visual observations and textual conclusions.

The following subsections examine four broad categories of reasoning mechanisms: chain-of-thought methods, program- and neuro-symbolic approaches, self-correction and reflection strategies, and interpretability-oriented analysis of reasoning processes. These mechanisms are not mutually exclusive; in practice, many recent systems combine several of them. Fig. 8 maps these mechanism families onto the overall multimodal reasoning pipeline.

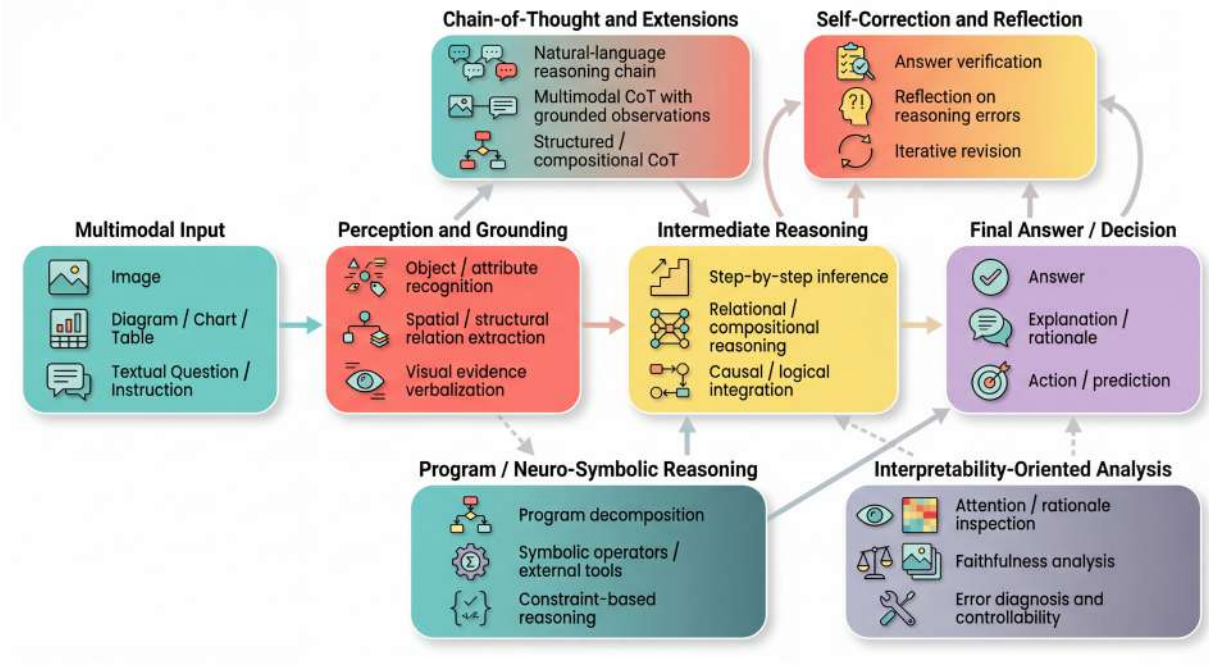


Figure 8: Overview of reasoning mechanisms and strategies in LVLMs for complex multimodal reasoning. The pipeline transforms multimodal inputs to answers via perception, intermediate inference, and generation. Four mechanisms intervene at different stages: chain-of-thought for stepwise decomposition, program-based methods for structured reasoning with external tools, self-correction for iterative refinement, and interpretability analysis for assessing faithfulness and robustness.

4.1 Chain-of-Thought and Its Extensions

Chain-of-thought (CoT) reasoning encourages models to generate explicit intermediate reasoning steps before producing a final answer [1,116]. Early studies demonstrate that providing intermediate reasoning steps can significantly improve performance across arithmetic, commonsense, and symbolic reasoning tasks [117,118]. In LVLMs, this idea has become one of the most influential mechanisms for improving multi-step inference, because it makes the reasoning trajectory more explicit and can partially reduce direct answer guessing based on superficial multimodal correlations.

For LVLMs, visual or multimodal CoT extends standard CoT by explicitly integrating visual observations into the reasoning chain [116,119]. A common pattern is to first identify salient visual evidence, then translate it into textual or semi-structured intermediate descriptions, and finally use these grounded observations to support subsequent inference. In this sense, multimodal CoT typically involves two coupled components.

Visual Grounding in Reasoning. The first component is the extraction or verbalization of visual evidence that is relevant to the reasoning process. This may take the form of textual descriptions of objects, attributes, spatial relations, charts, or diagram elements that can then be incorporated into the reasoning

chain [5,94]. The quality of this grounding step is critical: if the visual evidence is misidentified or inaccurately verbalized, subsequent reasoning may remain coherent in form while being unsupported by the input.

Multimodal Reasoning Chains. The second component is the explicit incorporation of grounded visual evidence into intermediate inference steps [36,116,120]. Rather than simply appending an image caption before answering, effective multimodal CoT requires the model to connect observations to relational, compositional, or causal inferences. For example, a valid chain may first identify that one object lies to the left of another and then use this grounded relation to answer a spatial query.

Representative methods such as Visual-CoT [119] and Multimodal-CoT/MMCoT [116,120] show that explicitly structuring reasoning around grounded visual evidence can improve performance on complex multimodal reasoning tasks. A common design choice in these approaches is to separate rationale generation from answer prediction, so that the model first constructs a reasoning trajectory and only then produces the final response. This decomposition can improve transparency and sometimes accuracy, although it does not by itself guarantee reasoning faithfulness.

CoT has also been extended in more structured directions. For instance, compositional chain-of-thought prompting [36] incorporates scene-graph-style relational structure into intermediate reasoning, thereby encouraging the model to reason over object relations more explicitly. More broadly, multimodal CoT methods increasingly attempt to move from free-form natural language rationales toward more structured intermediate representations.

Despite its practical effectiveness, CoT has clear limitations. First, generated reasoning chains may be plausible but unfaithful, reflecting post-hoc rationalization rather than the true basis of the model's prediction [59,60]. Notably, prior empirical studies have shown that even invalid or partially incorrect reasoning chains can still yield high task performance, suggesting that CoT may not faithfully reflect the underlying reasoning process [1,117,118]. Table 3 summarizes representative reasoning mechanisms in LVLMs, highlighting their core features and the main limitations observed in current studies.

Table 3: Representative reasoning mechanisms in LVLMs.

Mechanism	Representative Works	Core Feature	Observed Limitation
Multimodal CoT	Multimodal-CoT [116], Compositional Chain-of-Thought Prompting [36], Visual Chain of Thought [119], MM-CoT [120], Visual SKETCHPAD [58], Least-to-Most Prompting [121], Chain-of-Thought Prompting [1], Tree of Thoughts [57], EmbodiedVSR [74], VcCoT [10]	Explicitly externalizes intermediate reasoning steps with visual grounding	Verbose but potentially unfaithful reasoning
Program-of-Thought	PoT Prompting [43], ToRA [42], PAL [105], UniGeo [122], MathQA [123], Visual Programming [107], TheoremQA [28], GeoQA [81]	Converts reasoning into executable programs for verifiable intermediate states	Tool dependency and error propagation

(Continued)

Table 3 (continued)

Mechanism	Representative Works	Core Feature	Observed Limitation
Self-consistency/reflection	Self-Consistency [44], Masked Thought [124], Training Verifiers to Solve Math Word Problems [125], Math-Shepherd [126], WizardMath [17], DeepSeek-R1 [127], Reasoning-LM [38]	Aggregates multiple reasoning paths or iteratively refines intermediate outputs	Increased inference cost
Neuro-symbolic integration	LINC [62], NS-CL [63], NS-VQA [41], Neuro-VLMs [49], Neuro-Symbolic AI [51], Abstraction NNs [21]	Combines neural perception with symbolic constraints for structured reasoning	Symbol grounding challenges
Code-centric reasoning training	Code-Think [108], DeepSeek-R1 [127], Llemma [128], ToRA [42], PAL [105], PoT Prompting [43], OpenMathInstruct-2 [18], MAmmoTH2 [79]	Encourages structured intermediate computation via code-based reasoning	Limited generalization across domains

4.2 Program-of-Thought and Neuro-Symbolic Approaches

Program-of-thought (PoT) approaches reformulate reasoning problems as executable programs, typically using code interpreters or symbolic runtimes to carry out intermediate computations [43,105]. This idea is closely related to interpretable reasoning datasets such as MathQA [123], in which solving a problem involves generating an explicit sequence of operations rather than only a final answer. In multimodal settings, PoT is especially appealing because it offers a way to move part of the reasoning process from opaque neural inference into verifiable procedural execution. Fig. 9 contrasts implicit neural reasoning with program-of-thought and neuro-symbolic reasoning pipelines.

The key strength of PoT is that it externalizes reasoning into a structured form. Instead of relying solely on free-form natural language rationales, the model generates code or executable procedures whose intermediate states can be inspected and verified. This is particularly valuable for tasks involving arithmetic computation, symbolic manipulation, geometric reasoning, or other forms of structured inference in which correctness can be checked mechanically [43,105].

From a logical-reasoning perspective, PoT offers three major benefits. First, it makes reasoning **explicit**: each computational step is represented in a formal procedure rather than being hidden in model activations. Second, it adds **computational precision**: once the problem has been translated into an executable program, the system can rely on the external runtime for exact calculation or rule application. Third, it improves **debuggability**: errors can often be localized to specific steps in the generated program instead of being diffused across a natural-language rationale.

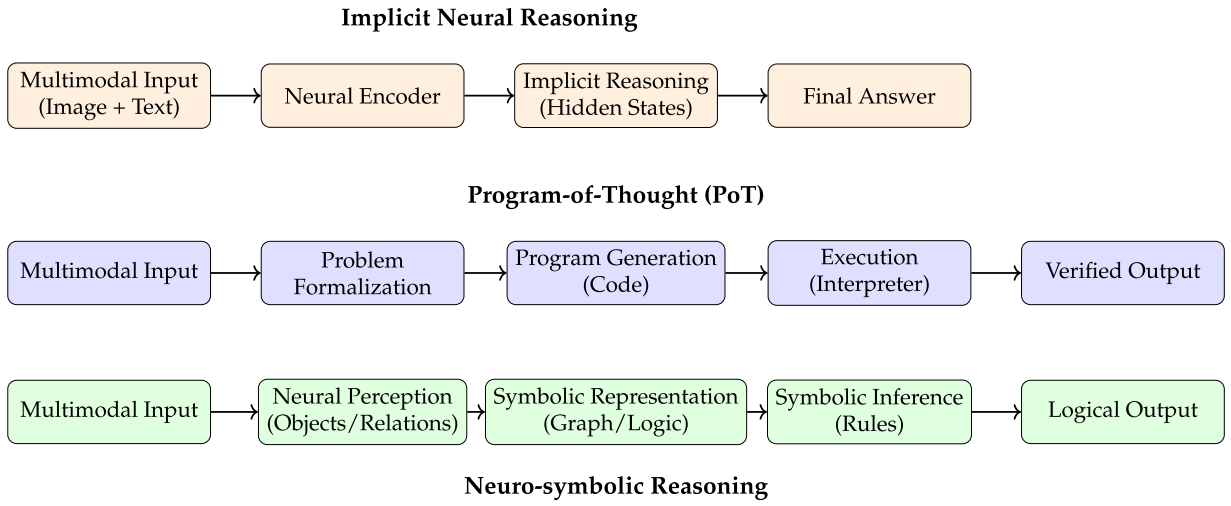


Figure 9: Program-of-thought and neuro-symbolic reasoning in LVLMs.Top: traditional neural reasoning relies on implicit hidden-state inference, which is difficult to verify. Middle: program-of-thought (PoT) converts reasoning into executable programs, enabling step-wise verification and precise computation. Bottom: neuro-symbolic approaches combine neural perception with symbolic inference, allowing explicit reasoning over structured representations.

Recent work on code-enhanced reasoning [38,108] further suggests that executable intermediate representations can improve reasoning beyond purely mathematical settings. However, the success of PoT depends critically on whether the model can correctly formalize the underlying problem. In multimodal contexts, this often requires translating visual inputs into structured symbolic variables, relations, or functions before execution. If this translation fails, the advantages of downstream executability are greatly reduced.

Neuro-symbolic approaches address a related problem from a complementary direction. Rather than merely converting reasoning into code, they explicitly combine neural perception modules with symbolic inference engines [41,63]. In these systems, neural components handle perception and language understanding, while symbolic components operate over structured intermediate representations to perform formal inference. This architecture is particularly well suited to tasks such as deductive reasoning, compositional visual reasoning, and logical entailment, where the reasoning process benefits from discrete structure and explicit rule application [62].

A representative example is Visual Programming (VisProg) [107], which decomposes complex visual tasks into modular executable procedures. More broadly, recent reviews of neuro-symbolic AI [51] and studies of robust reasoning in VLMs [49] show that neuro-symbolic methods remain one of the most promising routes toward stronger reasoning faithfulness and verifiability.

At the same time, both PoT and neuro-symbolic methods face a common bottleneck: the interface between neural representations and symbolic structure. To benefit from formal reasoning, the model must first produce accurate structured representations from ambiguous and noisy multimodal inputs. Errors in parsing, grounding, or formalization can easily propagate into the symbolic stage. For this reason, these approaches are often strongest on structured reasoning tasks, but less robust in open-ended scenarios where perceptual ambiguity and linguistic underspecification are substantial.

This point is crucial when comparing these methods with CoT-style reasoning. Program execution and symbolic inference can improve *step verifiability* once a correct formal representation has been obtained, but they do not automatically guarantee *reasoning faithfulness* in the broader multimodal sense. In practice, PoT often verifies the correctness of downstream computation rather than the correctness of upstream

visual formalization, while neuro-symbolic pipelines often verify rule application rather than the reliability of perceptual grounding. As a result, these methods reduce some forms of structured hallucination and arithmetic error, yet they do not fully eliminate the gap between answer correctness and genuinely grounded reasoning in open-ended LVLM settings [49,59,60]. Their main contribution is therefore best understood as relocating the faithfulness bottleneck from free-form generation to representation quality, grounding fidelity, and interface robustness.

4.3 Self-Correction and Reflection Mechanisms

Self-correction mechanisms aim to improve reasoning reliability by enabling models to detect, revise, or filter erroneous reasoning trajectories [44,57]. These mechanisms are particularly important for LVLMs because multimodal logical reasoning is vulnerable to two intertwined sources of error: perceptual mistakes in interpreting the input and inferential mistakes in chaining the intermediate steps. Without some form of internal correction or verification, such errors can propagate directly to the final answer.

One widely used strategy is **self-consistency** [44]. Instead of relying on a single reasoning chain, the model samples multiple reasoning paths and selects the answer that is most consistent across them. The intuition is that correct reasoning is more likely to converge to the same conclusion across diverse trajectories, whereas incorrect reasoning may produce unstable or conflicting outputs. In multimodal reasoning, this can reduce sensitivity to local reasoning errors or brittle prompt effects, although its effectiveness depends on the diversity and quality of the sampled chains.

A second strategy is **reflection** or **reflexive self-evaluation**, in which the model explicitly reviews its own reasoning before committing to a final answer [38]. Here, the model may be prompted to critique intermediate steps, reconsider missing evidence, or justify why the conclusion follows from the input. Reflection can be useful for identifying obvious inconsistencies or unsupported leaps, but it does not guarantee genuine error detection: a model may also rationalize an incorrect answer more confidently rather than correct it.

A third strategy is **iterative refinement**, where reasoning is updated across multiple rounds rather than produced in a single forward pass [57,121]. In these approaches, intermediate reasoning states are progressively revised, expanded, or reorganized. This is especially helpful for tasks requiring long reasoning chains or hierarchical decomposition, since early-stage drafts can be corrected before the final output is generated.

4.4 Interpretability of Reasoning Processes

Interpretability is crucial for understanding how LVLMs produce reasoning outputs and for diagnosing why they fail [59,60]. In the context of logical reasoning, interpretability is not merely a desirable auxiliary property; it is closely tied to the broader question of reasoning faithfulness. If a model produces a correct answer, we still need to understand whether the answer was derived through grounded inference, shortcut exploitation, or post-hoc explanation.

Several interpretability approaches are particularly relevant to multimodal logical reasoning.

Attention Visualization. One common approach is to inspect attention patterns to determine which visual elements and textual tokens influence the model during reasoning [88]. In multimodal settings, attention analysis can help reveal whether the model is attending to the relevant objects, regions, or relations when producing a reasoning trace or final answer. Although attention alone should not be overinterpreted as a full explanation of model behavior, it remains a useful diagnostic signal for detecting missing grounding, distractor sensitivity, or cross-modal misalignment.

Reasoning Step Tracking. Another important approach is to inspect intermediate reasoning states or generated step-by-step traces [120]. Tracking reasoning steps allows researchers to identify where a reasoning chain breaks down, whether the error originates from perception or inference, and how mistakes propagate across multiple steps. This is particularly valuable for tasks involving long-horizon reasoning, where final-answer evaluation alone provides little information about failure modes.

Explanation Analysis. A third approach is to examine generated natural-language explanations or rationales [1]. Such explanations can provide insight into the model's apparent reasoning strategy and may reveal whether the conclusion is supported by grounded evidence or instead reflects superficial pattern matching. However, explanation generation also has a major limitation: plausible explanations can be post-hoc and unfaithful. As a result, explanations should be treated as evidence for analysis, not as definitive proof of the model's internal reasoning process.

5 Benchmarks and Evaluation Protocols

Evaluating logical reasoning capabilities in LVLMs requires not only challenging benchmarks, but also evaluation protocols that can distinguish genuine reasoning from superficial answer matching. This section surveys representative benchmarks and evaluation practices, with a focus on how well current evaluation settings capture the diversity, depth, and faithfulness of complex logical reasoning [29,35].

As shown in Fig. 10, the evaluation of logical reasoning in LVLMs can be understood as a three-part framework. Existing benchmarks cover several recurring reasoning types, including inductive, compositional, deductive, and causal reasoning, but remain fragmented in coverage. On top of this benchmark landscape, evaluation protocols can be divided into answer-level and process-level metrics.

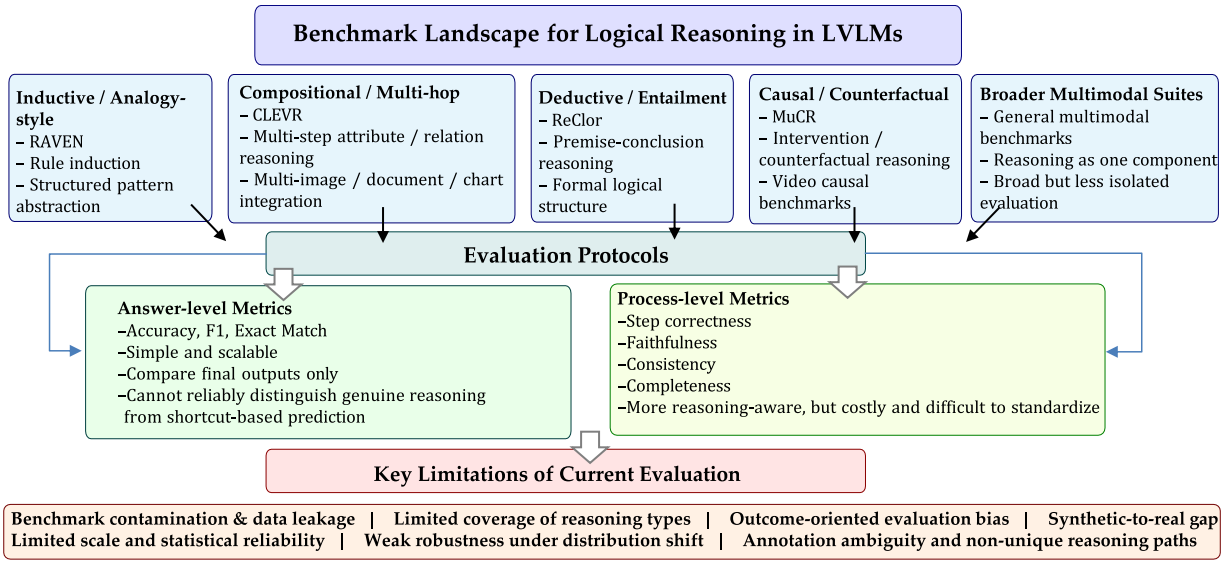


Figure 10: Framework view of benchmarks and evaluation protocols for logical reasoning in LVLMs, including both answer-level metrics and a structured set of process-level metrics (step correctness, faithfulness, consistency, completeness).

5.1 Existing Benchmarks for Logical Reasoning

We categorize benchmarks according to the primary form of reasoning they evaluate. This organization is not absolute, since many benchmarks involve multiple reasoning skills simultaneously. Nonetheless, it provides a useful overview of the current evaluation landscape and helps clarify which aspects of logical

reasoning are well covered and which remain underexplored. Table 4 provides a comparison of representative logical reasoning benchmarks, including their release year, primary reasoning type, modality, and core evaluation focus.

Table 4: Comparison of logical reasoning benchmarks.

Benchmark	Year	Reasoning Type	Modality	Core Focus
VQA [8]	2015	General visual reasoning	Vision + Language	Open-ended visual question answering over natural images
CLEVR [39]	2016	Multi-hop	Visual	Compositional relational reasoning
DocVQA [129]	2021	Document reasoning	Vision + Language	Visual question answering over document images with layout and OCR dependency
GeoQA [81]	2022	Geometric reasoning	Vision + Language	Diagram-based geometric reasoning with numerical inference
ChartQA [130]	2022	Mathematical reasoning	Vision + Language	Chart understanding with visual and numerical reasoning
UniGeo [122]	2022	Geometric reasoning	Vision + Language	Unified benchmark for diagram-based geometric logical and numerical reasoning
SEED-Bench [30]	2023	Multi-type	Vision + Language	Generative multimodal comprehension
MMBench [45]	2023	Multi-type	Vision + Language	Broad multimodal capability evaluation
MMMU [31]	2023	Expert-level reasoning	Vision + Language	Discipline-level question answering
TheoremQA [28]	2023	Mathematical reasoning	Vision + Language	Theorem-driven scientific and mathematical reasoning with formal knowledge dependence
MathVista [78]	2024	Mathematical reasoning	Vision + Language	Diagram and chart-based reasoning

(Continued)

Table 4 (continued)

Benchmark	Year	Reasoning Type	Modality	Core Focus
MMMU-Pro [31]	2024	Robust expert reasoning	Vision + Language	Advanced multimodal reasoning under increased difficulty
Video-MME [85,131]	2024	Temporal multi-hop	Video + Text	Comprehensive video reasoning evaluation
MuirBench [132]	2024	Multi-image reasoning	Multi-image + Text	Robust multi-image understanding
MathVerse [80]	2024	Mathematical reasoning	Vision + Language	Large-scale visual mathematical problem solving
OlympiadBench [27]	2024	Olympiad-level reasoning	Vision + Language	Challenging bilingual scientific problems requiring advanced multimodal reasoning
MME [29]	2024	Multi-type	Vision + Language	Broad evaluation suite for multimodal perception and reasoning abilities
SEED-Bench-2 [30]	2024	Multi-type	Vision + Language	Expanded multimodal benchmark with stronger coverage of reasoning and perception diversity
BRIDGE [55]	2024	Multi-hop	Document + Vision + Language	Multi-hop reasoning in long multimodal documents with grounded evidence
M3GQA [54]	2024	Multi-hop graph reasoning	Vision + Graph + Language	Multi-entity and multi-hop reasoning across graph-structured multimodal settings
MathWriting [133]	2025	Mathematical reasoning	Vision + Language	Handwritten mathematical expression understanding and reasoning
CausalVLBench [34]	2025	Causal reasoning	Vision + Language	Benchmarking causal attribution and intervention-oriented reasoning in LVLMs

(Continued)

Table 4 (continued)

Benchmark	Year	Reasoning Type	Modality	Core Focus
MuCR [10]	2025	Causal reasoning	Vision + Language	Multimodal causal link identification across visual and textual evidence
VcCoT [10]	2025	Causal chain-of-thought reasoning	Vision + Language	Evaluating causal reasoning with explicit multimodal reasoning chains
InPhyRe [67]	2025	Inductive physical reasoning	Vision + Language	Evaluates whether multimodal models can infer physical rules and abstract patterns from examples
DixitWorld [75]	2025	Abductive reasoning	Vision + Language	Evaluates plausible explanation generation through multi-agent Dixit-style gameplay
Mind the Gap [87]	2025	Spatial reasoning	Vision + Language	Comprehensive benchmarking of spatial reasoning failures and gaps in vision-language models
MM-IQ [46]	2025	Abstraction and reasoning	Vision + Language	Benchmarks human-like abstraction, analogy, and general reasoning in multimodal models
VCR-Bench [134]	2025	Video chain-of-thought reasoning	Video + Text	Comprehensive evaluation of explicit reasoning chains in video understanding

At a broad level, current benchmarks fall into several recurring groups: *inductive or analogy-style benchmarks*, *compositional and multi-hop benchmarks*, *deductive or entailment-oriented benchmarks*, and *causal or counterfactual benchmarks*. In addition, a number of broader multimodal evaluation suites include logical reasoning as one component, though often not as their exclusive focus.

Inductive reasoning benchmarks. RAVEN [22] is one of the most representative benchmarks for inductive reasoning in visual contexts. It evaluates rule induction through Raven-style matrix completion tasks, in which models must infer latent transformation rules from structured visual patterns. The benchmark includes multiple matrix configurations, such as relational and progressive structures, and therefore provides

a controlled testbed for evaluating whether a model can abstract patterns rather than merely recognize objects. Its main strengths lie in its clear focus on rule induction and its systematic control of difficulty. Its main limitation, however, is that it is primarily visual and synthetic, which constrains its representativeness for broader multimodal logical reasoning.

Compositional and multi-hop reasoning benchmarks. CLEVR [39] remains a foundational benchmark for compositional and multi-step visual reasoning. It presents synthetic scenes containing multiple objects and asks questions that require combining attributes, relations, and counting operations across several reasoning steps. CLEVR is valuable because it was explicitly designed to reduce dataset bias and isolate reasoning structure. However, its synthetic nature also limits ecological validity: the benchmark evaluates controlled compositional reasoning well, but does not capture the ambiguity and noise of real-world multimodal inputs.

Beyond synthetic scenes, more recent multi-hop settings attempt to evaluate more realistic evidence integration. Such benchmarks are motivated by the observation that many real reasoning tasks require models to connect information scattered across different regions, images, or modalities rather than answer from a single local cue [54–56]. This broader class of benchmarks is particularly important for evaluating logical reasoning in document, chart, and multi-image settings.

Deductive reasoning benchmarks. ReClor [64] is a representative benchmark for deductive reasoning and logical entailment. It presents reading comprehension problems that require identifying whether a conclusion follows from a set of premises through valid logical inference. Its main value lies in its emphasis on formal reasoning structure rather than open-ended commonsense interpretation. However, because it is fundamentally text-based, its relevance to multimodal LVLM evaluation is indirect: it is highly useful for conceptual grounding, but does not by itself test whether a model can integrate visual evidence into deductive inference.

Causal reasoning benchmarks. Causal reasoning has only recently begun to receive dedicated multimodal benchmark support. MuCR (Multimodal Causal Reasoning Benchmark) [10] is a representative example. It evaluates whether models can identify causal relations across visual and textual evidence, distinguish correlation from causation, and reason about interventions or counterfactual changes. Compared with earlier general multimodal benchmarks, MuCR is notable because it explicitly targets a class of reasoning that is central to real-world decision making but difficult to capture with answer-only evaluation.

Related efforts extend causal evaluation to temporal settings. Video-based causal benchmarks assess whether models can infer causal relations from event sequences rather than static images alone [34]. Such tasks are especially important because causal reasoning often depends on temporal structure, yet they are also more difficult to evaluate reliably due to longer contexts and greater annotation complexity.

Broader observations on current benchmark coverage. Taken together, current benchmarks cover important but still fragmented slices of the reasoning problem. RAVEN captures abstract pattern induction, CLEVR captures synthetic compositional reasoning, ReClor captures text-based deductive reasoning, and MuCR and related datasets begin to probe multimodal causal inference. However, few benchmarks simultaneously combine real-world multimodal complexity, diverse reasoning types, and process-aware evaluation. This fragmentation remains one of the central challenges in assessing logical reasoning in LVLMs [25,29].

It is also worth noting that large-scale resources not specifically designed for logical reasoning have still played an enabling role in benchmark development and model training. For example, YouTube-8M [86] provided large-scale video infrastructure that later supported progress in video-language understanding and temporal reasoning. Although such resources do not directly evaluate logical inference, they contributed to the broader ecosystem in which reasoning-oriented multimodal systems emerged.

5.2 Evaluation Metrics

Evaluating complex logical reasoning in LVLMs remains fundamentally challenging because correct answers do not necessarily imply valid reasoning processes [125,126]. Existing evaluation protocols can be broadly divided into *answer-level* metrics and *process-level* metrics, each capturing a different aspect of model behavior. Fig. 11 summarizes this distinction between answer-level and process-level evaluation.

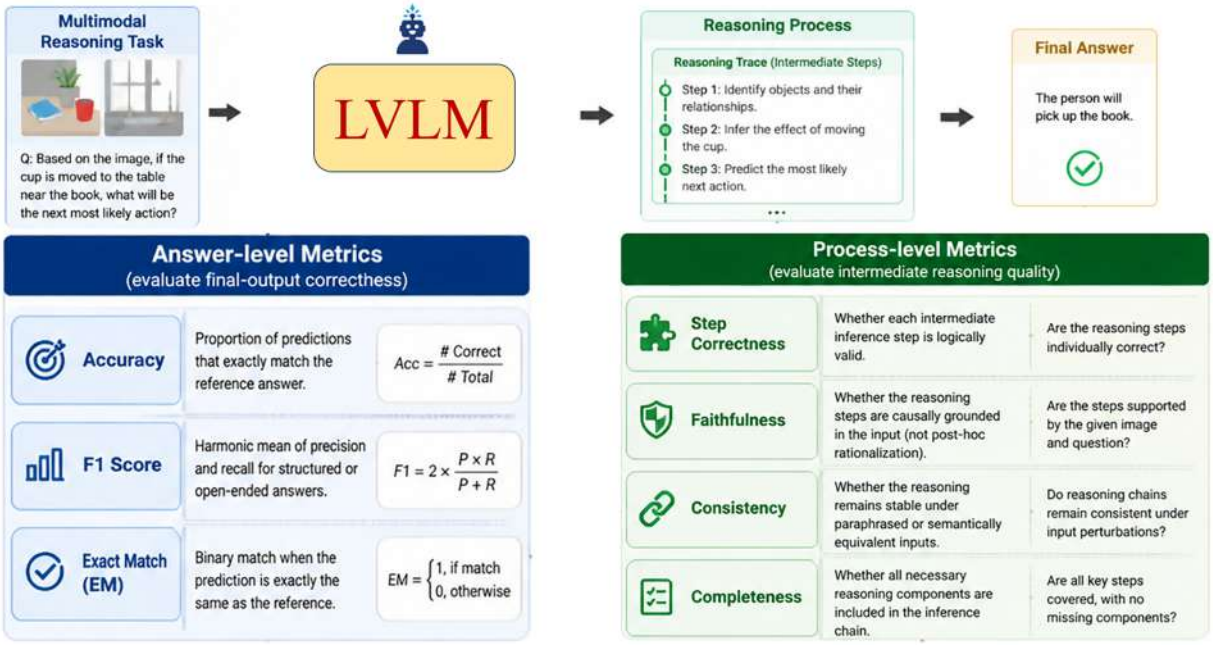


Figure 11: Evaluation metrics for complex logical reasoning in LVLMs, including Answer-level metrics and Process-level metrics. However, process-level evaluation is harder to scale because it often requires annotated reasoning traces, verifier models, or controlled perturbation tests.

Answer-level metrics. Answer-level metrics evaluate whether the final prediction matches a reference answer, without explicitly considering how that answer was obtained. Common examples include *accuracy*, *F1 score*, and *exact match*. These metrics remain widely used because they are simple, scalable, and easy to compare across models and benchmarks.

However, their limitations are especially pronounced in logical reasoning tasks. First, answer-level scores cannot distinguish between *genuine inference* and *shortcut-based prediction*. A model may produce the correct answer by exploiting dataset regularities or shallow multimodal cues rather than by carrying out the intended reasoning process. Second, answer-only evaluation cannot represent the existence of multiple valid reasoning paths, which is common in open-ended or multi-step tasks. Third, strict surface-form matching can be brittle: two answers may be semantically equivalent while differing in wording or decomposition.

Process-level metrics. To address these weaknesses, process-level metrics attempt to evaluate the quality, validity, and structure of intermediate reasoning steps [124,126].

These dimensions can be organized as a structured evaluation protocol:

1. *Step correctness*: verifying whether each intermediate inference step is logically valid;
2. *Faithfulness*: assessing whether reasoning steps are causally grounded in the input rather than post-hoc rationalization;
3. *Consistency*: testing robustness under paraphrased or semantically equivalent inputs;

4. *Completeness*: ensuring that all necessary reasoning components are included in the inference chain.

Despite their conceptual appeal, process-level metrics are difficult to deploy at scale. They often require detailed annotations of intermediate steps or access to internal model traces, both of which are expensive and difficult to standardize across tasks [125,126]. Moreover, ground-truth reasoning trajectories are often not unique: different but equally valid solution paths may exist for the same problem [124]. This makes process supervision and evaluation substantially harder than answer-only scoring.

More fundamentally, even current process-level metrics do not fully capture several key aspects of multimodal logical reasoning, including cross-modal grounding validity, abstraction consistency, and causal coherence [34,35]. For this reason, there remains a strong need for evaluation frameworks that jointly assess *correctness*, *faithfulness*, and *generalization* under realistic multimodal conditions [29,46].

To make the discussion of empirical findings more explicit, we further summarize the main patterns observed across existing evaluations. Current LVLMs generally show stronger performance on perception-oriented and short-answer multimodal tasks, but their performance becomes less stable when tasks require multi-step inference, cross-modal evidence integration, counterfactual reasoning, or faithful explanation generation. Therefore, we emphasize not only reported performance scores but also the evaluation conditions, reasoning requirements, and failure patterns behind these results.

5.3 Cross-Method Quantitative Comparison and Standardized Evaluation Criteria

Although existing studies report substantial progress in LVLM reasoning, direct cross-method comparison remains difficult because different works adopt different benchmarks, answer formats, decoding settings, and modality assumptions [2,3,50]. To make the comparison more systematic, we further conduct a standardized quantitative synthesis over representative LVLMs and reasoning-oriented benchmarks. Rather than treating heterogeneous benchmark results as fully interchangeable, we organize the comparison along three reasoning-relevant axes: (1) single-image multimodal reasoning, measured by MathVista [78], MMBench [45], and MMMU [31]; (2) multi-image reasoning, measured by MuirBench [132]; and (3) video temporal reasoning, measured by Video-MME [131]. These benchmarks are selected because they are widely used in recent LVLM evaluation and provide explicit quantitative results for representative models [4,30].

We adopt the following standardization criteria. First, we only include numerical results that are explicitly reported in the original papers or official benchmark pages [7,37,95]. Missing results are marked as "–" and are not imputed. Second, all reported values are accuracy scores in percentage form unless otherwise noted. Third, for the single-image reasoning axis, we compute a simple descriptive average over MathVista, MMBench, and MMMU only when all three values are available [19,26].

$$\text{R-Avg}_{\text{SI}}(m) = \frac{1}{3}\left(A_{\text{MathVista}}(m) + A_{\text{MMBench}}(m) + A_{\text{MMMU}}(m)\right), \tag{1}$$

where $A_b(m)$ denotes the accuracy of model m on benchmark b. This quantity is intended only as a coarse descriptive index, not as a statistically rigorous meta-analytic estimate, because MathVista, MMBench, and MMMU differ in domain mix, answer format, and evaluation protocol. Fourth, for Video-MME, we report the "without subtitles" setting when both subtitle-free and subtitle-enhanced results are available, because it better isolates visual-temporal reasoning from additional textual transcript information. Finally, we do not collapse single-image, multi-image, and video scores into a single overall score, since these axes measure different reasoning conditions and should not be interpreted as strictly equivalent. Open-weight and proprietary systems are also listed separately in Table 5, because their training data, system prompts, and evaluation interfaces are not fully standardized.

Table 5: Standardized quantitative comparison of representative LVLMs on reasoning-relevant benchmarks. All numbers are accuracy scores (%). "–" indicates that no directly comparable value was found in the surveyed sources. R-Avg$_{SI}$ is used only as a descriptive single-image summary when MathVista, MMBench, and MMMU are all reported. Open-weight and proprietary models are listed separately because their system boundaries and training conditions are not fully standardized. For Video-MME, we report the subtitle-free setting.

Model/Method	Main Design Paradigm	MathVista	MMBench	MMMU	R-Avg$_{SI}$	MuirBench	Video-MME
Open-weight/publicly described systems							
Qwen-VL-Max [7]	Unified LVLM/visual-language alignment	51.0	77.6	51.4	60.0	–	–
InternVL2-8B [95]	Scaled vision backbone and alignment	58.3	81.7	49.3	63.1	–	–
InternVL2-26B [95]	Scaled vision backbone and alignment	59.4	83.4	48.3	63.7	–	–
LLaVA-OneVision-0.5B [100]	Unified multi-scenario transfer	34.8	52.1	31.4	39.4	25.5	44.0
LLaVA-OneVision-7B [100]	Unified multi-scenario transfer	63.2	80.8	48.8	64.3	41.8	58.2
LLaVA-OneVision-72B [100]	Unified multi-scenario transfer	67.5	85.9	56.8	70.1	54.8	66.2
Proprietary systems							
GPT-4V [37]	Proprietary LVLM	49.9	75.0	56.8	60.6	62.3	59.9
GPT-4o [37]	Proprietary omni-modal LVLM	63.8	–	69.1	–	–	71.9

Table 5 and Fig. 12 summarize the resulting comparison. The results show several recurring trends. First, model scaling improves reasoning performance, as seen from the LLaVA-OneVision series: R-Avg$_{SI}$ increases from 39.4 for the 0.5B model to 70.1 for the 72B model, while MuirBench [132] and Video-MME [131] also improve consistently. Second, strong single-image reasoning does not automatically imply equally strong multi-image or video reasoning. For example, GPT-4V obtains a competitive single-image descriptive average [37,47], but LLaVA-OneVision-72B [100] reports stronger Video-MME performance in the subtitle-free setting. Third, the multi-image setting remains particularly challenging: even strong models show a noticeable drop on MuirBench compared with standard single-image benchmarks such as MMBench [45] and MMMU [31]. These observations should be interpreted within access-matched groups or as broad evidence trends rather than as a fully controlled ranking across all open and proprietary systems.

The quantitative comparison also clarifies the limitations of current evidence. Some reasoning mechanisms reviewed in Section 4, such as program-of-thought [43], neuro-symbolic reasoning [49,51], and self-correction [44], often report results on task-specific benchmarks rather than on a shared multimodal reasoning suite. For instance, MathVista [78] reports tool-augmented textual baselines such as PoT GPT-4 with caption and OCR inputs, but these results are not directly equivalent to end-to-end LVLM inference because the visual information has already been converted into text [26]. Therefore, instead of claiming a universal ranking across all reasoning mechanisms, we treat Table 5 as a conservative evidence map: it compares representative systems only where public and directly comparable benchmark scores are available. This analysis reinforces the need for future standardized evaluation protocols that jointly report

final accuracy, reasoning-process validity, modality-specific grounding, and robustness under distribution shift [13,19,25].

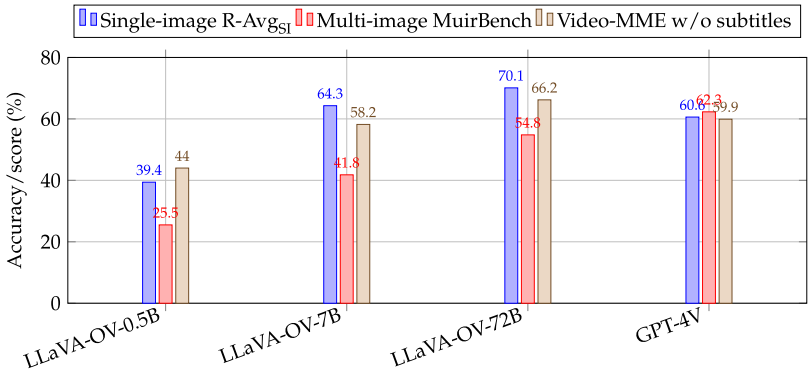


Figure 12: Multi-axis quantitative comparison of representative LVLMs. Single-image R-Avg$_{SI}$ is a descriptive average of MathVista, MMBench, and MMMU when all three are reported. MuirBench evaluates robust multi-image reasoning, while Video-MME evaluates video understanding under the subtitle-free setting. The comparison is intended as an evidence map rather than a fully controlled ranking, and it shows that improvements on single-image reasoning do not necessarily transfer uniformly to multi-image and video reasoning.

5.4 Limitations of Current Evaluation

Despite the growing number of multimodal benchmarks, existing evaluation protocols remain insufficient for rigorously assessing complex logical reasoning in LVLMs [25,29]. Fig. 13 summarizes the major limitations of current LVLM evaluation protocols. Existing benchmarks mainly emphasize final-answer accuracy, but this metric alone cannot reveal whether the model follows a valid reasoning process, grounds its answer in visual evidence, or remains robust under distribution shift and adversarial perturbations.

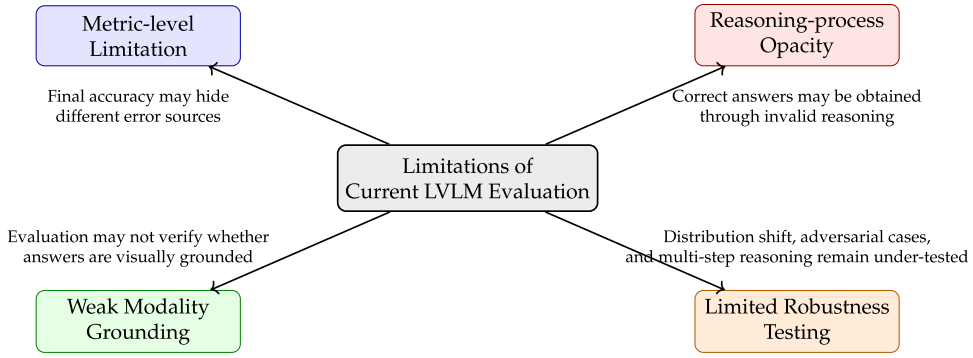


Figure 13: Main limitations of current LVLM evaluation protocols. Existing benchmarks often emphasize final-answer accuracy, while providing limited evidence about reasoning-process validity, visual grounding, and robustness under challenging conditions.

Benchmark contamination and data leakage. A fundamental concern is that benchmark instances may appear, directly or indirectly, in large pretraining corpora [37]. Because many popular benchmarks are publicly available, large-scale models may partially memorize patterns or examples from them, which can inflate apparent reasoning performance. This is especially problematic for logical reasoning, where benchmark success may otherwise be misinterpreted as evidence of genuine inferential capability [135].

Limited coverage of reasoning types. Most existing benchmarks probe only a narrow subset of reasoning skills, such as visual pattern induction [22] or synthetic compositional reasoning [39]. Yet real multimodal reasoning often requires the joint use of multiple forms of inference, including deductive, abductive, causal, and compositional reasoning [9,23,75]. The fragmented nature of current benchmark design therefore makes comprehensive evaluation difficult.

Outcome-oriented evaluation bias. A dominant limitation of current practice is the heavy reliance on final-answer correctness [44]. This biases evaluation toward outcomes rather than reasoning validity. As a result, models that exploit superficial multimodal regularities may appear competitive with models that genuinely perform structured inference. Existing benchmarks are therefore often insufficient to separate *true reasoning* from *shortcut-based inference* [1].

Synthetic-to-real gap. A substantial portion of existing reasoning benchmarks is based on synthetic data [22,39]. Synthetic settings are useful because they provide control and reduce annotation ambiguity, but they lack the noise, ambiguity, and long-tail variability of real-world multimodal inputs. Models that perform well in controlled synthetic environments may therefore fail to transfer their reasoning ability to realistic scenarios [31].

Scale and statistical reliability. Many logical reasoning benchmarks are relatively small, limiting their power to support statistically stable model comparison [30,45]. Small test sets are more vulnerable to variance, narrow coverage, and benchmark-specific artifacts, making it difficult to draw reliable conclusions about general reasoning ability.

Distributional robustness and generalization. Performance on standard benchmarks often degrades substantially under distribution shift, including changes in visual style, wording, task format, or evidence layout [31,46]. This raises a broader concern: even when a model performs well on a benchmark, it may still lack the robustness required for practical reasoning deployment.

Annotation quality and ambiguity. Evaluation is further complicated by annotation noise, underspecification, and the fact that many logical reasoning tasks admit multiple plausible solution paths [126]. In such cases, defining a single gold answer or a single gold reasoning trace may be inherently reductive, which complicates fair comparison across systems.

6 Challenges and Open Problems

Rather than repeating earlier descriptive comparisons, this section distills the main unresolved *technical bottlenecks* that recur across architectures, mechanisms, datasets, and evaluation settings. We focus on five cross-cutting issues: process validity, long-horizon state management, multimodal grounding, data quality, and evaluation realism. The goal is not to restate every earlier limitation, but to synthesize the recurring failure patterns that most consistently prevent current LVLMs from becoming reliable reasoning systems.

6.1 Reasoning Faithfulness and Validity

A central challenge is ensuring that model outputs reflect genuine logical inference rather than superficial pattern matching [1,44]. Even when LVLMs produce correct answers, those answers may be driven by spurious correlations, memorized shortcuts, or dataset regularities rather than valid reasoning [135].

This faithfulness problem has several sources. First, training data may contain exploitable biases, allowing models to succeed without learning the intended reasoning patterns [33]. Second, current benchmarks often emphasize final-answer accuracy and therefore do not clearly distinguish between genuine inference and shallow answer prediction [29]. Third, many reasoning tasks involve implicit premises, background

assumptions, or underspecified constraints, making it difficult to determine whether a model is reasoning correctly or simply generating plausible outputs [62].

Recent work on neuro-symbolic reasoning and reasoning robustness has further emphasized that answer correctness alone is insufficient evidence of valid reasoning [49]. Addressing this challenge requires training objectives and evaluation protocols that target faithfulness directly, rather than treating it as a by-product of answer accuracy.

6.2 Hallucination in Reasoning

Hallucination remains a major challenge in LVLM reasoning, but in reasoning settings it is especially problematic because errors can compound across inference steps [59,60]. A hallucinated intermediate conclusion may appear coherent and even support later steps, ultimately leading to a convincing but incorrect final answer.

This problem is more difficult than ordinary generation errors because reasoning hallucinations are often structurally plausible. Models may invent unsupported relations, attributes, or causal links, and such errors can remain hidden inside a multi-step chain [90]. The risk increases when reasoning chains become longer, when external grounding is weak, or when the model must infer latent structure from noisy visual evidence [44,57].

Reducing reasoning hallucination therefore requires more than improving fluent generation. It requires stronger grounding, intermediate verification, and mechanisms for checking whether each step is supported by the input or by trusted external knowledge [82].

6.3 Long-Range Reasoning Capabilities

Complex logical reasoning often requires maintaining coherence across long inference chains, yet long-range reasoning remains a persistent weakness of current models [57,121]. As the number of reasoning steps increases, models must preserve relevant intermediate states, retrieve earlier premises, and prevent local errors from cascading through later steps.

Current LVLMs still struggle with these requirements. Long reasoning chains place pressure on memory, attention allocation, and step-by-step consistency [32]. Earlier mistakes can easily propagate, and models may lose track of previously established constraints or fail to integrate them correctly into later stages [44,116]. These issues are particularly severe in multi-image, long-document, and video settings, where evidence is distributed across extended multimodal contexts.

Improving long-range reasoning will likely require better memory mechanisms, more reliable intermediate-state management, and architectures or training strategies that explicitly support long-horizon inference.

6.4 Cross-Modal Representation Alignment

Robust logical reasoning in LVLMs depends on accurate alignment between visual and textual representations [5,52]. This remains difficult because the two modalities differ substantially in form: visual information is continuous, spatial, and often ambiguous, whereas textual information is discrete, sequential, and usually more explicit [7].

For reasoning tasks, alignment must go beyond coarse semantic matching. Models must determine which visual evidence corresponds to which textual premise, maintain consistency across modalities, and resolve conflicts when the two sources provide incomplete or partially inconsistent information [45,111].

Misalignment at this stage can undermine the entire reasoning process, even if the downstream inference mechanism is otherwise strong [95].

This challenge suggests that future work should not treat representation alignment as merely a perception problem. Instead, alignment mechanisms must be designed with reasoning in mind, especially for tasks involving compositional structure, abstraction, and multimodal constraint satisfaction.

6.5 Data Quality and Scarcity

The development of logical reasoning in LVLMs is constrained not only by limited data volume, but also by the quality of the reasoning data itself [12,26]. In contrast to general multimodal instruction data, reasoning-oriented corpora must often include carefully designed problem structures, valid intermediate reasoning steps, reliable annotations of latent relations, and enough diversity to prevent shortcut learning. These requirements make high-quality data expensive to construct and difficult to scale.

This challenge has several dimensions. First, annotating reasoning trajectories requires expertise and is often labor-intensive [64]. Second, synthetic data can help scale reasoning supervision, but may not transfer well to realistic multimodal settings [22,39]. Third, dataset quality is highly sensitive to hidden sampling bias, annotation regularity, and source imbalance. Seemingly broad datasets may still overrepresent particular domains, visual layouts, answer formats, or reasoning templates, which can allow models to exploit superficial regularities instead of learning transferable inference patterns [31,39,45]. In multimodal reasoning, such bias may appear as shortcut dependence on OCR artifacts, visual co-occurrence priors, or benchmark-specific answer distributions.

Data quality problems also interact with evaluation quality. If benchmark construction reuses narrow templates, weak negative cases, or partially contaminated sources, answer accuracy can overestimate genuine reasoning ability [29,48]. For this reason, better reasoning data should be judged not only by scale, but also by diversity, annotation reliability, source balance, and resistance to shortcut exploitation.

As a result, progress in reasoning often depends on relatively small or specialized datasets, which is unlikely to be sufficient for building broad, robust, and transferable multimodal reasoning capabilities. Better data construction pipelines, scalable synthetic-real mixtures, stronger annotation audits, and higher-quality process supervision remain important open problems.

6.6 Evaluation Gaps

Finally, current benchmarks and evaluation protocols still provide only a partial view of reasoning ability [29,134]. Benchmark performance may not reliably reflect whether a model can reason in a grounded, generalizable, and faithful manner [31,45].

Several issues contribute to this gap. Existing benchmarks often cover only a limited subset of reasoning types [35]; many are susceptible to contamination or data leakage [48]; and most still emphasize answer-level outcomes over reasoning processes [120]. As discussed in the previous section, this makes it difficult to distinguish true reasoning from benchmark-specific shortcuts.

For this reason, evaluation remains one of the most important unresolved problems in the field. Progress will depend on benchmarks that cover a broader range of reasoning skills, are more robust to contamination and distribution shift, and include process-aware protocols capable of assessing not only correctness, but also faithfulness and reasoning quality.

Taken together, these bottlenecks show that the central obstacle is no longer raw benchmark coverage alone, but the difficulty of maintaining grounded, verifiable, and transferable reasoning under realistic

multimodal conditions. Table 6 summarizes the main challenge-to-evidence mapping discussed in this section and highlights the most actionable open directions [24,25].

Table 6: Challenge-to-evidence mapping in recent LVLM reasoning research.

Challenge	Representative Evidence	Main Observation	Open Direction
Faithfulness gap	Neuro-symbolic reliability analyses [49,51], Multimodal CoT studies [116,120]	Correct final answers can still arise from brittle shortcuts, implicit heuristics, or post-hoc rationalizations rather than truly grounded intermediate reasoning	Process-grounded supervision, verifier-guided decoding, and causally faithful reasoning traces
Evaluation saturation	Harder benchmarks such as MMMU-Pro and MuirBench [31,132], broader benchmark analyses [25,29]	Many earlier benchmarks are becoming less discriminative, making it difficult to separate shallow answer matching from robust multimodal reasoning ability	Continually updated hard-split benchmarks, adversarial benchmark construction, and process-sensitive evaluation
Temporal reasoning bottleneck	Video-MME [131], VCR-Bench [134], video QA benchmarks such as TVQA and ActivityNet-QA [83,84]	LVLMs remain substantially weaker on long-horizon temporal dependency modeling than on static image reasoning, especially when evidence must be aggregated across distant frames	Memory-aware temporal reasoning, event-structured representations, and long-context video planning mechanisms
Math-visual reasoning instability	MathVista [78], MathVerse [80], OlympiadBench [27], UniGeo [122]	Models frequently fail in symbolic precision, formula grounding, diagram interpretation, and multi-step numerical consistency even when coarse semantic understanding is correct	Hybrid symbolic tool use, theorem-aware solvers, and explicit visual-to-formal representation interfaces
Cross-scene aggregation weakness	LLaVA-OneVision [100], MuirBench [132], LLaVA-NeXT-Interleave [72]	Evidence fusion across multiple images or interleaved contexts is markedly less stable than single-image inference, with errors in entity alignment and cross-image comparison	Structured multi-image planning, entity-centric memory, and graph-based cross-scene reasoning

(Continued)

Table 6 (continued)

Challenge	Representative Evidence	Main Observation	Open Direction
Spatial reasoning fragility	Mind the Gap [87], Think3D [115], EmbodiedVSR [74], 3D-R1 [136]	Spatial relations, viewpoint changes, occlusion, depth cues, and 3D consistency remain persistent failure points even for otherwise strong general-purpose LVLMs	Geometry-aware intermediate representations, 3D-augmented supervision, and embodied spatial reasoning frameworks
Causal reasoning deficiency	CausalVLBench [34], Multimodal Causal Reasoning Benchmark/MuCR/VcCoT [10], CausalBench [35]	Current models often capture correlation patterns rather than genuine causal structure, leading to weak performance on intervention, counterfactual, and cause-effect attribution tasks	Interventional data construction, causal representation learning, and counterfactual multimodal reasoning pipelines
Inductive generalization weakness	RAVEN [22], InPhyRe [67], MM-IQ [46], inductive multimodal learning studies [66]	Models still struggle to abstract reusable rules from limited observations, particularly when test distributions differ from training patterns or require physical/general relational induction	Rule-centric pretraining, abstraction-oriented curricula, and out-of-distribution inductive evaluation
Abductive reasoning uncertainty	Reasoner [23], DixitWorld [75], MAR [9]	Inferring the most plausible hidden explanation from incomplete multimodal evidence remains difficult, especially when several explanations are superficially compatible	Hypothesis ranking objectives, uncertainty-aware decoding, and explanation-calibrated training
Grounding and hallucination tension	Hallucination analyses such as DHCP and CLAIM [59,60], broad LVLM evaluations [30,45]	Strong fluent generation can mask weak visual grounding, causing unsupported object mentions, attribute errors, or multilingual hallucinations during reasoning	Grounding-aware decoding, cross-modal verification, and uncertainty-calibrated generation

(Continued)

Table 6 (continued)

Challenge	Representative Evidence	Main Observation	Open Direction
Document and chart reasoning robustness	BRIDGE [55], ChartQA [130], DocVQA [129], WebSRC [137]	Performance drops substantially when reasoning depends on layout structure, dense OCR, long documents, or mixed visual-textual evidence distributed across pages or regions	Layout-aware memory, OCR-reasoning co-optimization, and hierarchical document planning
Tool-use reliability and execution brittleness	Chameleon [71], Toolformer [106], ToRA [42], PAL [105]	Tool-augmented reasoning can improve difficult tasks, but end-to-end reliability is limited by planning errors, incorrect tool calls, execution failures, and cascading mistakes	Tool-selection policies, execution verification, and adaptive planner-verifier architectures
Reasoning cost and scalability	InternVL2.5 [101], GLM-4.1V-Thinking/GLM-4.5V [102], Vision-R1 [70], WeThink [68]	Stronger reasoning often depends on larger models, longer chains, test-time scaling, or reinforcement learning, increasing training and inference cost substantially	Efficient reasoning distillation, adaptive compute allocation, and budget-aware test-time scaling
Benchmark-to-reality transfer gap	Survey and benchmark analyses [24,25,29], WebArena [91]	Gains on curated benchmarks do not always translate to open-world multimodal tasks, where perception noise, ambiguity, tool interaction, and long-tail cases are much more severe	Realistic interactive benchmarks, deployment-oriented evaluation, and robustness testing under distribution shift

7 Future Directions

Despite rapid progress, complex logical reasoning in LVLMs remains far from being reliably solved. To avoid overly general recommendations, we organize future research directions around concrete and actionable technical routes, including structured reasoning data construction, reasoning-centric architecture design, verifier-guided training, process-aware evaluation, and deployment-oriented robustness testing [24,25]. Fig. 14 organizes these future directions into five interconnected research dimensions.

Data: Constructing Structured Reasoning Corpora with Grounded Process Supervision. Future progress will depend heavily on better reasoning-oriented data. Existing multimodal datasets are still limited in realism, diversity, and process supervision [31,45]. A key direction is therefore to construct datasets that more faithfully reflect real-world reasoning conditions, including ambiguity, incomplete information, noisy evidence, and cross-modal inconsistency. Another important direction is to move from answer-only supervision toward richer annotations of intermediate reasoning steps, grounded evidence, and alternative valid solution paths [32,120]. Synthetic data will likely remain important for scaling, but it should be

combined with realistic distributions and better data curation to avoid overfitting to artificial patterns [39]. Curriculum-style data construction, in which reasoning complexity increases progressively, may further support the acquisition of compositional and long-horizon reasoning abilities [121].

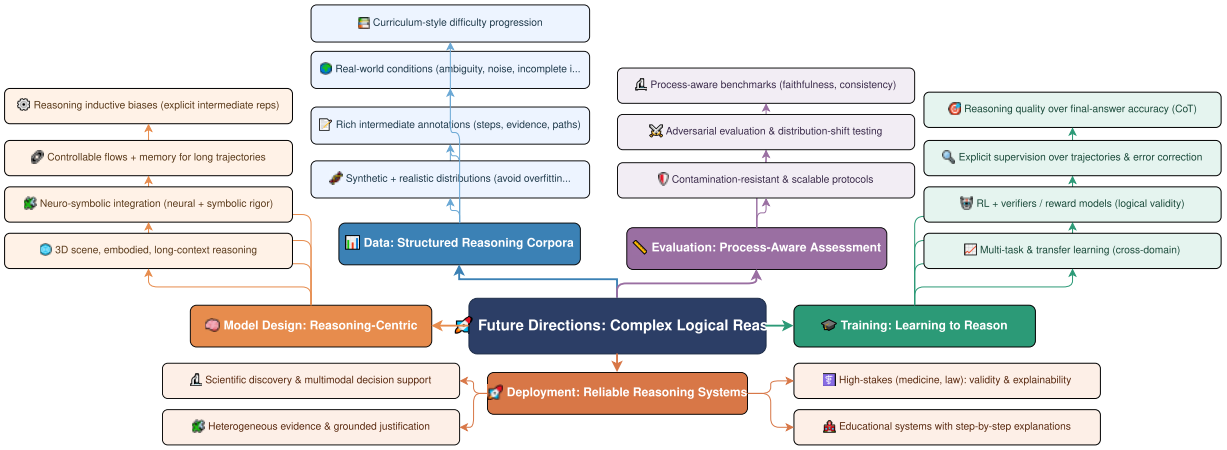


Figure 14: Overview of future research directions for complex logical reasoning in LVLMs. We organize future opportunities and progress into five interconnected dimensions: data, model design, training, evaluation, and applications/deployment.

Model Design: Toward Reasoning-Centric Architectures. Most current LVLMs are still optimized primarily for multimodal perception, generation, and instruction following, rather than for structured reasoning itself [5,95]. A more actionable direction is to design LVLMs with explicit intermediate-state memory, modular perception-reasoning interfaces, and controllable reasoning flows, allowing models to store premises, update intermediate conclusions, and verify cross-modal consistency during multi-step inference [41,63,138]. Neuro-symbolic integration remains especially promising because it offers a potential balance between neural flexibility and symbolic rigor [51,62]. More broadly, future architectures may need to support not only static 2D understanding, but also richer settings such as long-context multimodal reasoning, embodied interaction, and 3D scene reasoning [136]. At the same time, architectural advances must remain computationally practical, since many current reasoning-oriented systems rely on expensive inference-time procedures [57].

Training: Learning to Reason, Not Just to Predict. Training paradigms should incorporate explicit supervision and optimization signals for reasoning quality, including step-level correctness, evidence grounding, contradiction detection, and verifier-based reward signals, rather than relying only on final-answer accuracy [1,44]. While reasoning-focused fine-tuning can improve performance, future progress will likely require more explicit supervision over intermediate trajectories, grounded evidence usage, and error correction behavior [116]. Reinforcement learning is a particularly promising direction when combined with verifiers, reward models, or structured feedback that assess logical validity rather than only task success [125,127]. In addition, multi-task and transfer learning may help reasoning capabilities generalize across domains rather than remaining benchmark-specific [79]. A further underexplored direction is training-time diagnosis of recurrent reasoning failures, so that models can be improved through targeted correction rather than only broader scaling [126].

Evaluation: Beyond Accuracy toward Process-Aware Assessment. Evaluation will remain a central bottleneck unless it becomes more process-aware and more robust to shortcut solutions [29,134]. Future evaluation protocols should report not only final accuracy, but also step correctness, evidence faithfulness,

cross-modal grounding consistency, robustness under paraphrased or perturbed inputs, and contamination-resistant performance on newly constructed test splits [120]. Adversarial evaluation, distribution-shift testing, and contamination-resistant benchmark design will be essential for measuring whether models truly generalize [34,35]. Another important direction is the development of scalable evaluation protocols that can assess reasoning processes without requiring prohibitively expensive human annotations. More broadly, future evaluation should better reflect the properties that matter in practice: interpretability, robustness, and reliable multimodal inference under realistic conditions [29,119].

Applications and Deployment: From Controlled Tasks to Reliable Reasoning Systems. Ultimately, the value of logical reasoning in LVLMs should be evaluated under deployment-oriented conditions, such as noisy visual inputs, incomplete textual context, conflicting multimodal evidence, tool-use failures, and distribution shifts [12]. Promising directions include scientific discovery, multimodal decision support, educational systems with step-by-step explanations, and high-stakes domains such as medicine and law, where reasoning validity and explainability are especially important [15]. These application settings place stricter demands on reliability, transparency, and robustness than controlled benchmarks do [25]. They also highlight the need for models that can reason over heterogeneous evidence, justify their conclusions in grounded ways, and remain stable under uncertainty. In this sense, real-world deployment should not be viewed merely as an application endpoint, but also as an important driver of future research priorities.

8 Conclusion

In this survey, we present a structured and comprehensive synthesis of complex logical reasoning in LVLMs, covering conceptual foundations, architectural paradigms, reasoning mechanisms, evaluation protocols, and empirical findings. Our analysis highlights a fundamental tension in current LVLM research: although these models achieve strong performance on benchmark tasks, their reasoning processes are often implicit, fragile, and insufficiently grounded in multimodal evidence. Moreover, this survey identifies several key challenges that continue to limit progress, including insufficient reasoning faithfulness, difficulties in long-horizon inference, imperfect cross-modal alignment, and the lack of process-aware evaluation frameworks. Looking ahead, we argue that future research should move beyond performance-driven optimization and focus more explicitly on reasoning-oriented design. In addition, there is a clear need for more rigorous and scalable evaluation methodologies that can assess reasoning faithfulness, robustness, and generalization. Overall, we believe that addressing these challenges will enable LVLMs to move beyond strong perceptual performance toward reliable, interpretable, and generalizable multimodal reasoning systems. We hope this survey can provide both a conceptual foundation and a practical roadmap for advancing research in this important direction.

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